

IT&C Security Master

- www.ism.ase.ro -

Academy of Economic Studies

Bucharest, 2025

presentation

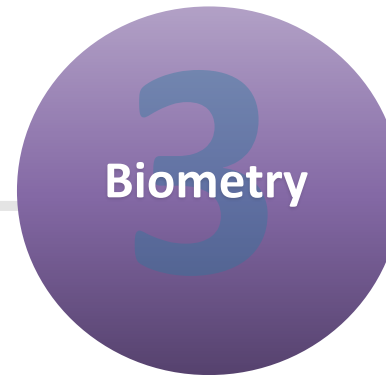
Multimedia Security & Biometrics

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Agenda for presentation





Structure, Organization Units, Evaluation

Course Management & Overview



1. Course Management & Overview

Multimedia Security & Biometrics

- Website:

www.ism.ase.ro

www.portal.ism.ase.ro

- Period:

May 17-18th, 2025



Multimedia Security – Objective

To provide a comprehensive understanding of multimedia content, its structure, and how security threats affect multimedia data. Students will learn techniques to inspect and secure multimedia content at the byte level, focusing on protection against common security vulnerabilities such as unauthorized access, data tampering, and malware.

1. Course Management & Overview

Multimedia Security – Evaluation

- When ? June, 2025
- Where ? in campus
- How ? on Quiz Online
- What ? Questions on Multimedia Security & Biometrics, Multiple Answers / Multiple Choices
theoretical and practical questions
- How long ? 60 min. quiz;

1. Course Management & Overview

Multimedia Security & Biometrics Evaluation

- 40% - Practical Evaluation
 - 10% - E1(17.05) – Steganography;
 - 10% - E2(18.05) – Biometry;
 - G1(19.05) – Exercise1 – Group1;
 - G2(20.05) – Exercise2 – Group2;
 - 20% Evaluation (23.05) – E1 & E2;
- 60% - QE

1. Course Management & Overview

Multimedia Security – Course Agenda

1. Image Standards

- BMP – internal representation, technical manipulation, image processing algorithms

2. Audio Formats

- RIFF / WAVE – byte level access, audio representation, audio processing algorithms

3. Video Content

- Representation, frame access, codecs, interpolation

4. Security concerns

- Steganography, digital watermarking, steganalysis, multimedia extraction and interpretation, security characteristics, digital rights management, content digital signature

1. Course Management & Overview

Multimedia Security – Bibliography

- [1] *Perspectives on Multimedia: Communication, Media and Information Technology* – R. Burnett, A. Brunstrom, A. Nilsson, Editura Wiley, 2004, 250 pg., ISBN 9780470868638
- [2] *Multimedia: concepte si practica* – I. Smeureanu, G. Drula, Ed. Cison, 1997, ISBN 973-96370-8-6
- [3] *Multimedia: Making it Work* – T. Vaughan, Sixth Edition, McGraw-Hill Publisher, 2004
- [4] *Intelligent Multimedia Communication: Techniques and Applications* – C. We Chen, Z. Li, S. Lian, Editura Springer, 2011, 300 pg., ISBN 9783642116858
- [5] *Multimedia Content and the Semantic Web: Standards, Methods and Tools* – G. Stamou, S. Kollias, Editura Wiley, 2005, 414 pg., ISBN 9780470857533
- [6] *Compression for Multimedia* – I. Bocharova, Cambridge University Press, 2010, 280 pg., ISBN 9780521114325

Section Conclusions

Security aspects applied in multimedia field is needed because of a high degree of malware found in any branch of digital content.





Images, Sounds, Videos, Digital Signature, Steganography, Watermarking, DRM

Multimedia Security Aspects



2.1 Security in Multimedia

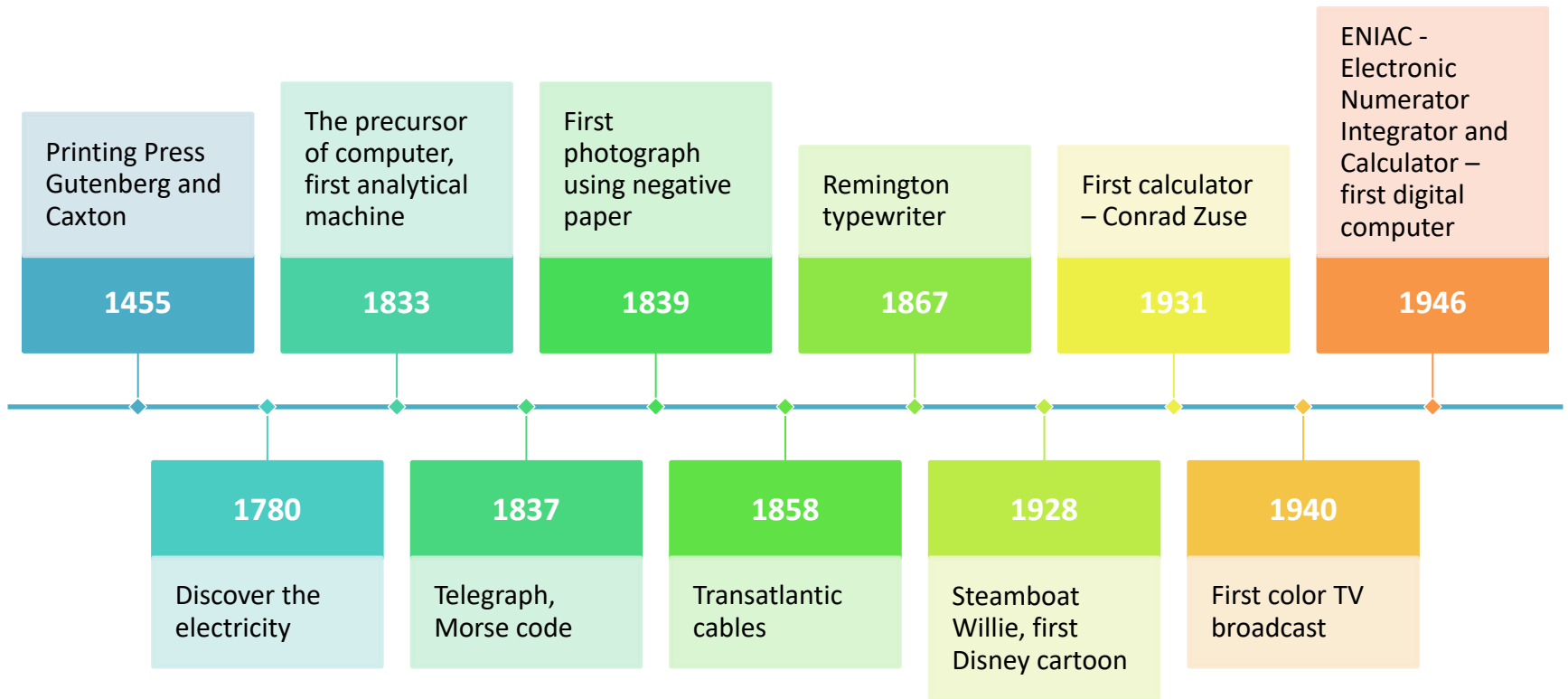
What is **M**ultimedia **S**ecurity – **MS** ?

A combination of text, digital graphics, sound, animation and video powered by a computer or by any electronic or digital means.

An instrument used to protect against unwanted or accidental actions of malicious users.



2.1 History



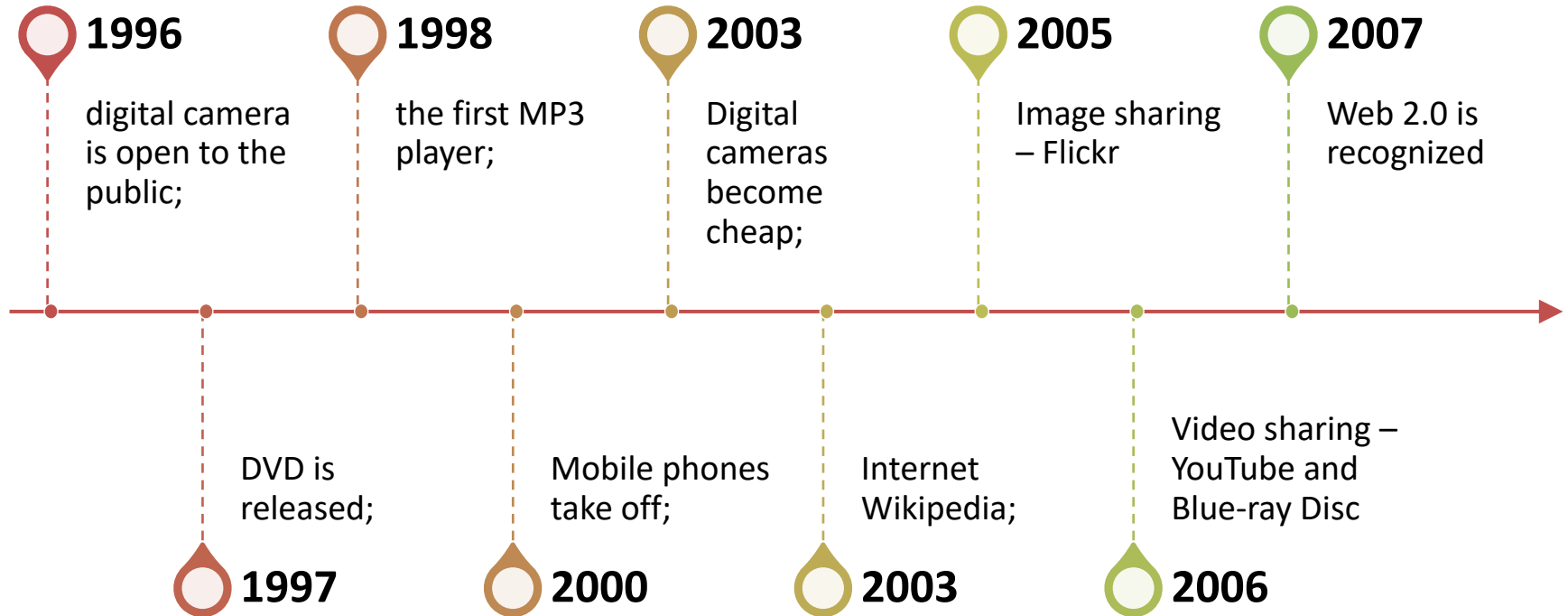
2.1 Security in Multimedia

- 1959 – Transistors replace vacuum tubes;
- 1965 – the optical fiber standards;
- **1970 – birth of The Internet;**
- 1971 – first email sent;
- 1977 – Apple II, first PC with colour interface;
- 1981 – first IBM PC;
- 1983 – the CD (compact disc) is announced;
- 1984 – ARPANET, the actual version Internet;
- **1985 – Commodore Amiga, the first multimedia PC;**
- 1985 – CD-ROM (CD read-only memory) is released;
- **1986 – *The Academic American Encyclopedia* on CD;**

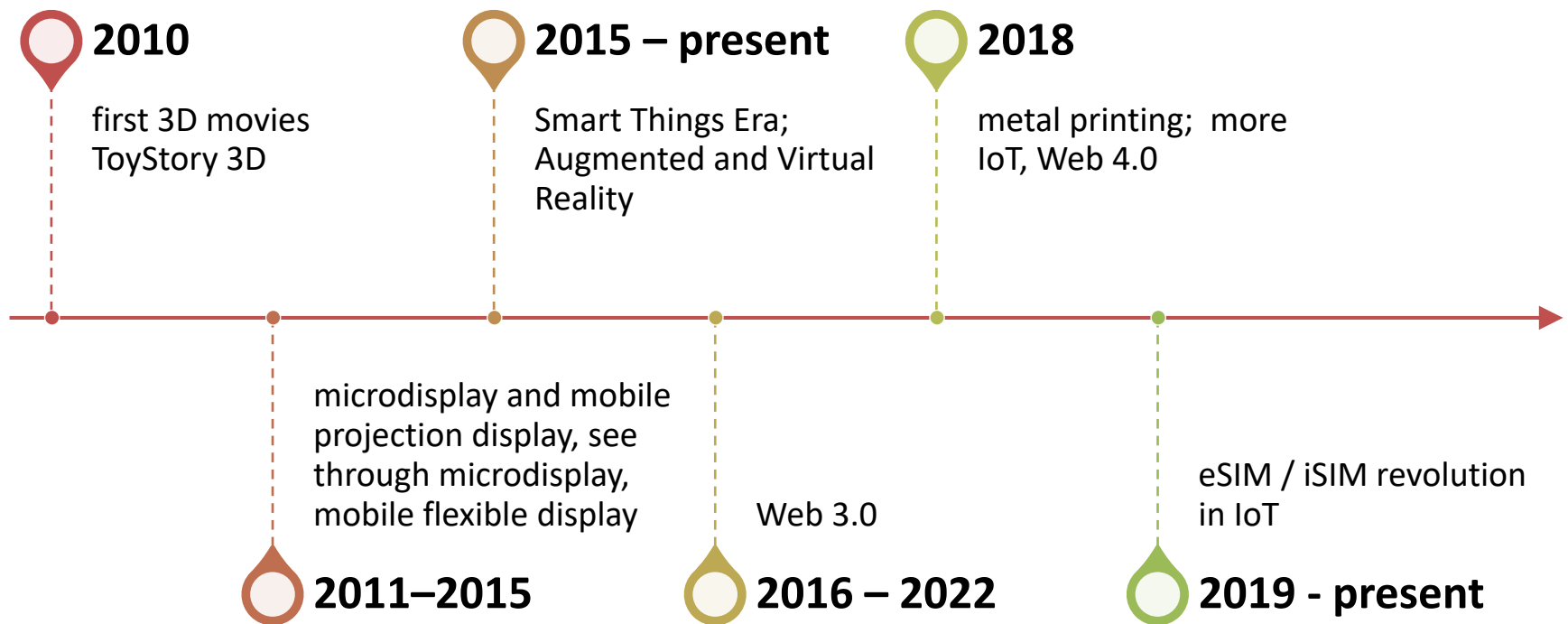
2.1 Security in Multimedia

- **1988** – Macromind launches Director, an instrument for *multimedia authoring*.
- **1991** – World Wide Web is launched;
- **1991** – MP3 format is created;
- **1993** – Mosaic, first Graphic Web browser;
- **1995** – Disney launches *Toy Story*, first movie created entirely on computer (77 minutes = 4 years + 800,000 hours for graphics processing);
- **1995** – Internet becomes broader;

2.1 Security in Multimedia



2.1 Security in Multimedia



2.1 Security in Multimedia

By target:

- Interactive (PTP – person to person)
 - Voice (voice)
 - Text (sms)
 - Image(mms)
 - Video (p2p calls)
 - Complex (social networking)
- Non-interactive (PTC – person to content)
 - text / pictures
 - internet
 - photos
 - video, music, movies

2.1 Security in Multimedia

By domain:

- Training (educational)
- Medical (Tomography, Echograph, RMN)
- Industrial (Simulation of a technological industrial flow)
- Geographic (Digital Maps)
- Services, Sales, Advertising

By destination and interactivity:

- Public and personal interest
- Local and remote
- Videoconference

2.1 Security in Multimedia

Peripheral devices and hardware components:

- Image: scanner, digital camera, film scanner, mobile device;
- Sound: sound board (analogous to digital convertor), microphone;
- Video: capture board, video camera, TV tuner;

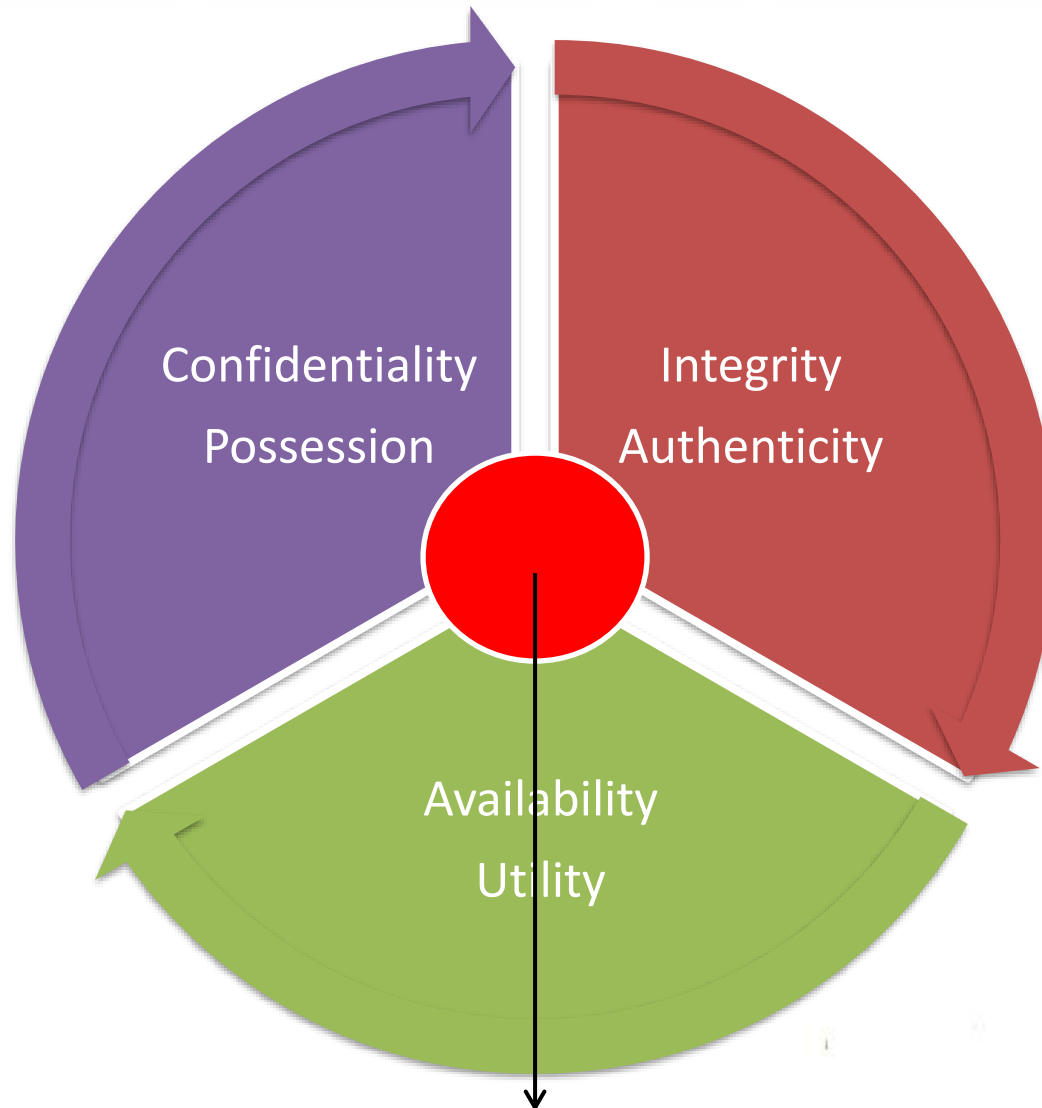
2.1 Security in Multimedia

Drivers to access multimedia hardware components;

Specialized software for multimedia processing;

Specialized software for integrating security aspects;

2.1 Security in Multimedia



Multimedia Security Aspects

2.1 Security in Multimedia

Why do we need **MS** for ?

- **Watermarking and authenticity**
 - Digital watermarking
 - Image self-restoration
- **Steganography and confidentiality**
 - Embedding messages in multimedia files
 - Identifying altered digital content
- **Multimedia forensic and integrity**
 - Image and Video forensic
- **Usable Security**
 - CAPTCHA, Graphical Passwords, FIDO, Biometry

Is what you expected?

This was [Section 2.1 – Security in Multimedia](#)

For non-repudiation:

Digital Signature

For protection:

Digital Rights Management

For confidentiality:

Steganography

For authenticity:

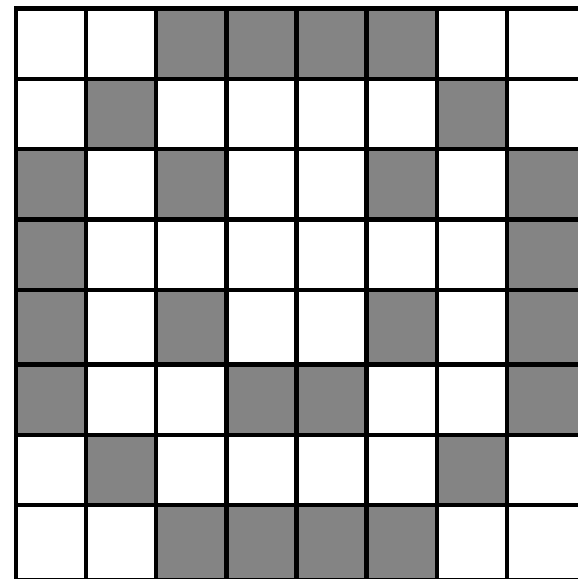
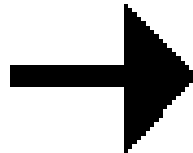
Watermarking

For integrity:

Watermarking and Steganography

2.2 Images

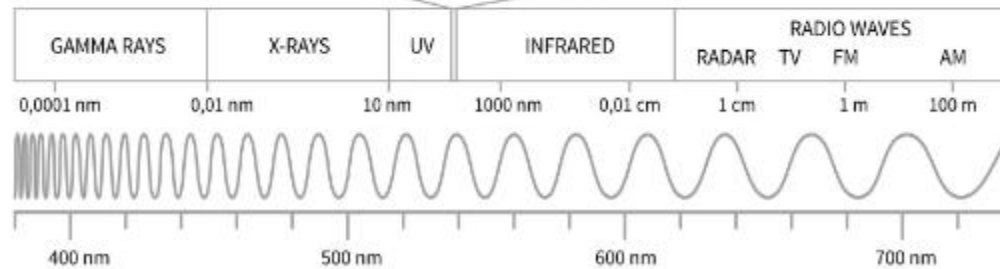
00111100
01000010
10100101
10000001
10100101
10011001
01000010
00111100



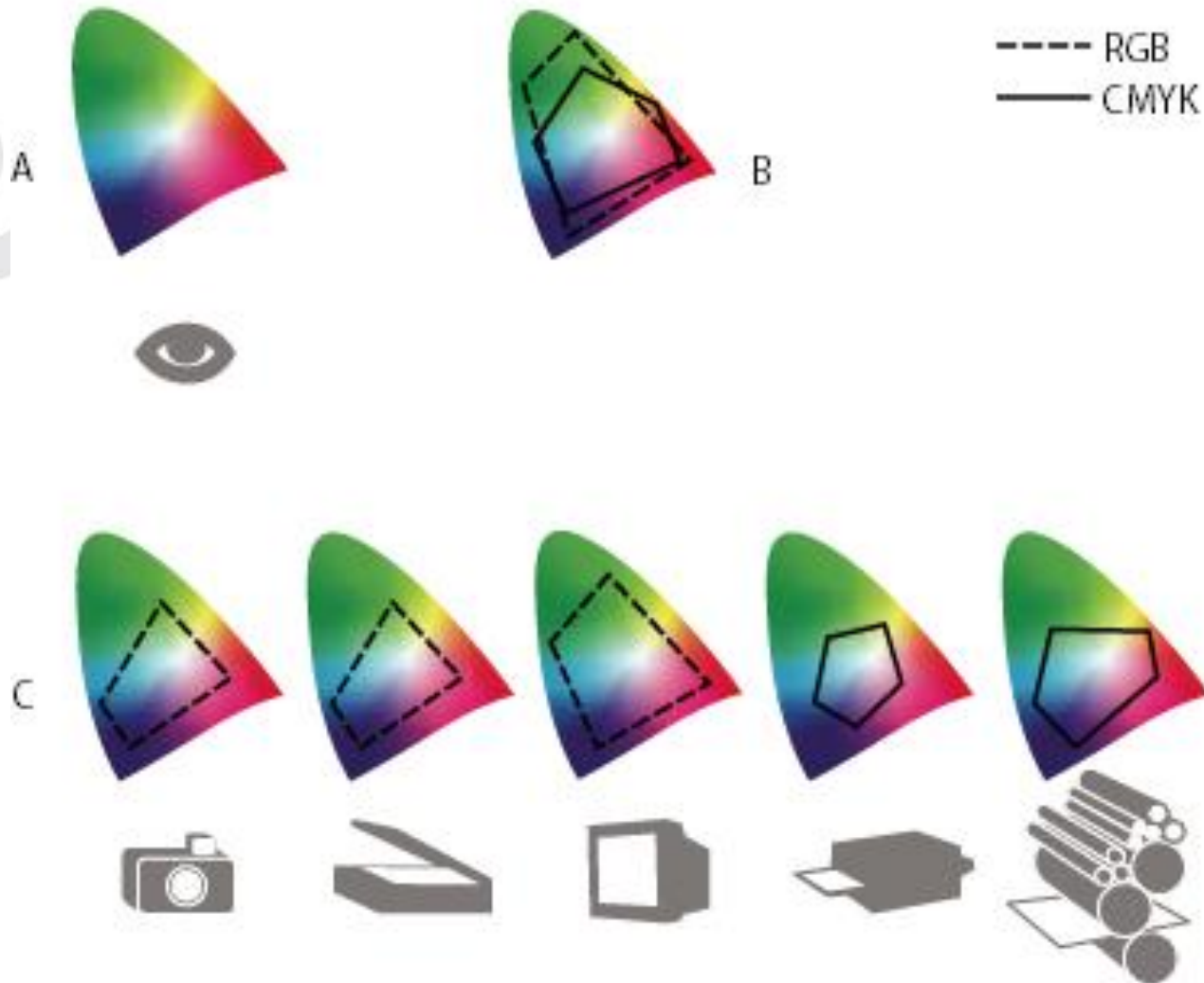
VISIBLE SPECTRUM



VISIBLE LIGHT



Image



Types of color palettes:

A – CIE Lab

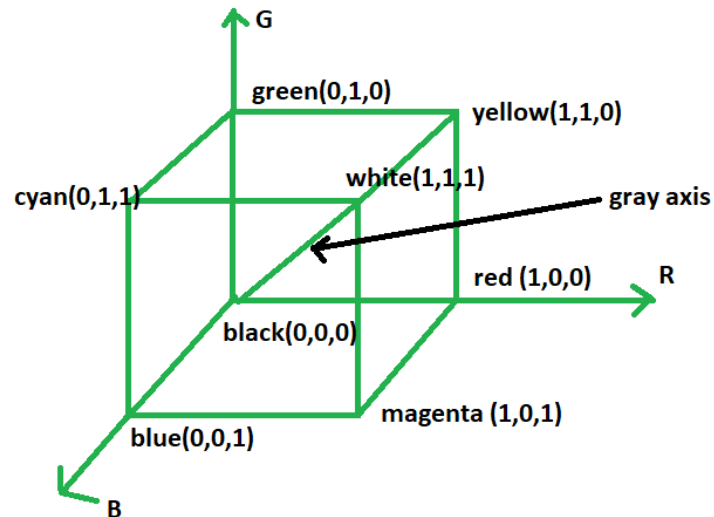
B – Digital images

C – Devices

Image

RGB

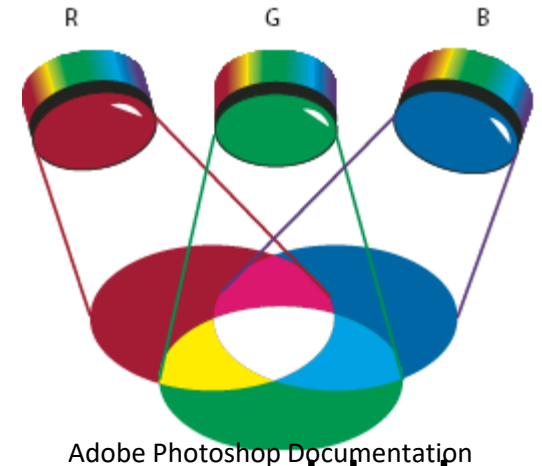
- uses 3 main colors: Red, Green, Blue
- each color with an intensity between 0 and 255
- for equal values: $R=G=B$, gray tones are created



Image

RGB

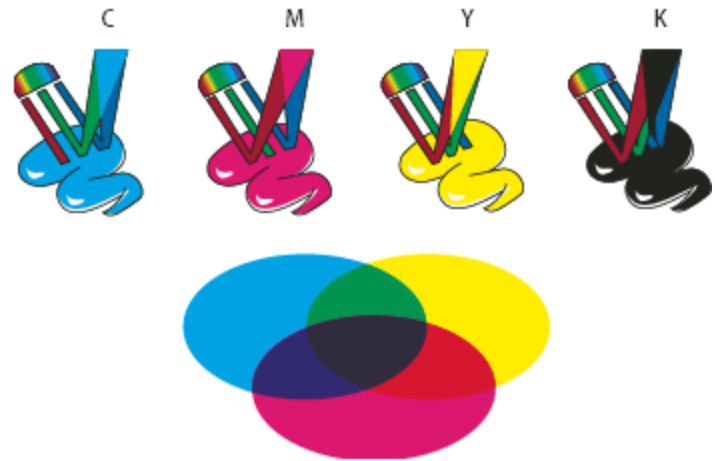
- additive colors
- the absence of the 3 main colors generates black (0 for each color)
- the presence of all 3 colors at maximum intensity generates white
- different color pallets: Adobe RGB, sRGB, ProPhoto RGB
- used for displaying devices



Image

CMYK

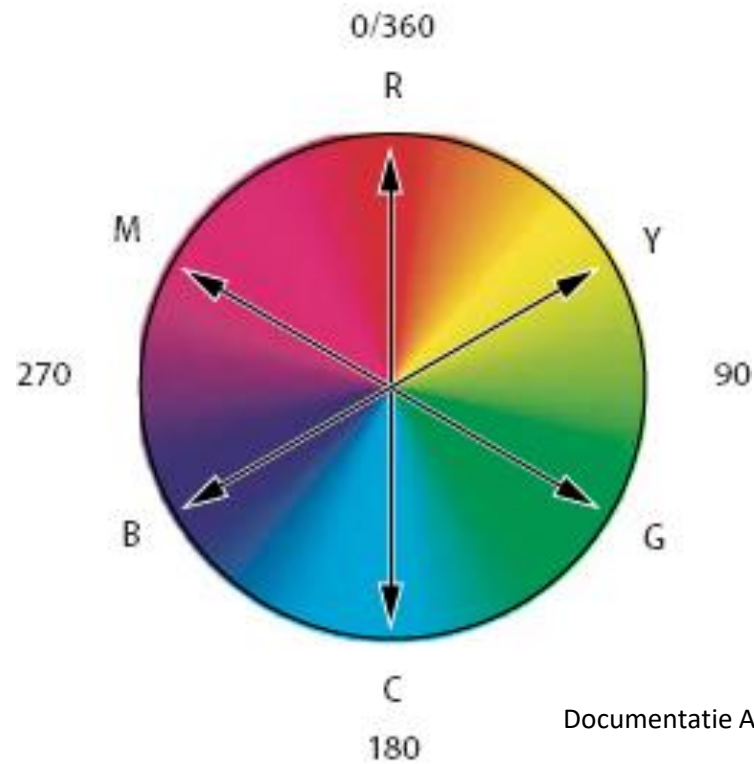
- Uses four based colors (Cyan, Magenta, Yellow, Black)
- Each color with an intensity between 0% and 100%
- Are subtractive colors
- Used for printing devices



Adobe Photoshop Documentation

Image

RGB vs CMYK



Documentatie Adobe Photoshop

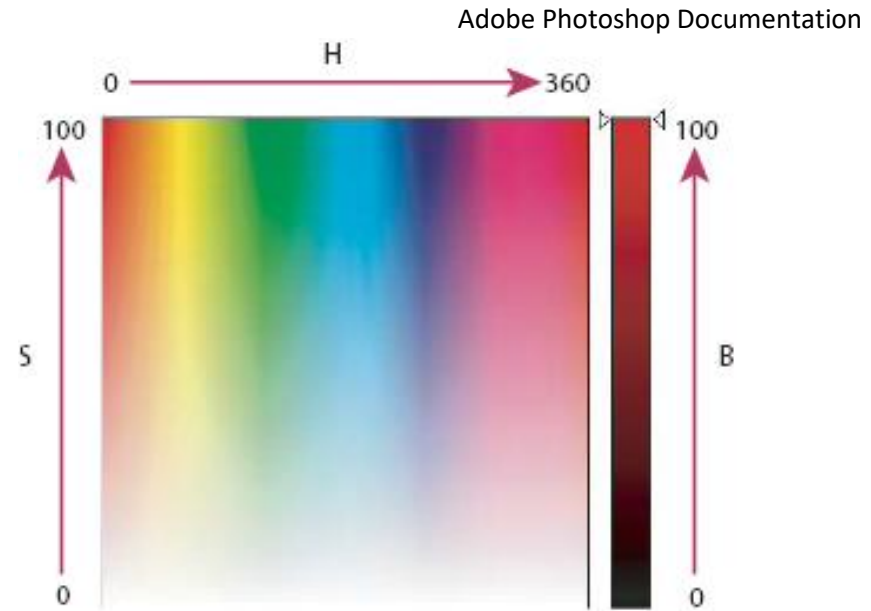
RGB to CMYK

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix} \quad \begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Image

HSB/HSV

- Model based on the Hue, Saturation, Brightness / Lightness properties
- Based on the human perception of colors



HSB Model

H – Hue
S – Saturation
B - Brightness

Image

Lab Color

- defined also as CIE L^*a^*b (Lightness, a – green-red component, b – blue-yellow component);
- L is between 0 and 100, a and b are between +127 and – 128;
- based on the human perception of colors
- the numeric values from this model cover all the colors perceptible by the human eye;
- it's a device independent model;

Image

Grayscale

- Defines grayscale tones;
- In an image with a 8 bits depth there are 256 gray tones;
- Gray tones can be represented as black percentages (0% - 100%)
- Each gray pixel has a intensity between 0, black and 255 white;

Image

Web-safe

- A limited set of colors derived from RGB
- Uses 256 colors;
- The color set is used by all browser;

Image

Hue

Brightness

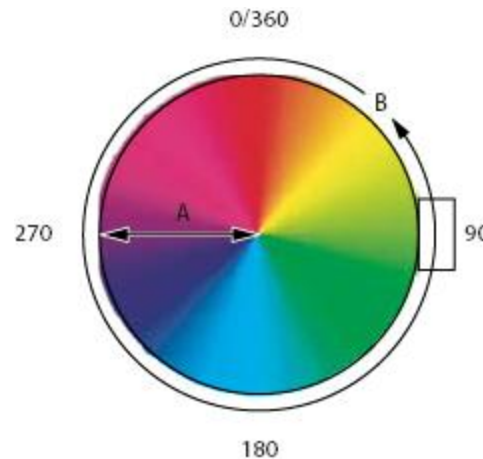
Saturation

Contrast

Color balance

Image

Hue is the level of color reflected by an object and transmitted to the human eye;



Color pallets:
A – Saturation
B – Hue

Image

01

Saturation

- Describe the intensity of a color;
- Represents the level of gray color correlated with the hue, with 0% (gray) 100% (maximum intensity)

02

Brightness

- High and low intensity measured from 0 % (black) to 100% (white)

Image

Contrast

- Measures the difference between brightest and darkest areas

Color balance

- Analyses the color pallet distribution
- The predominant color influences all other colors that are present in the picture



Image

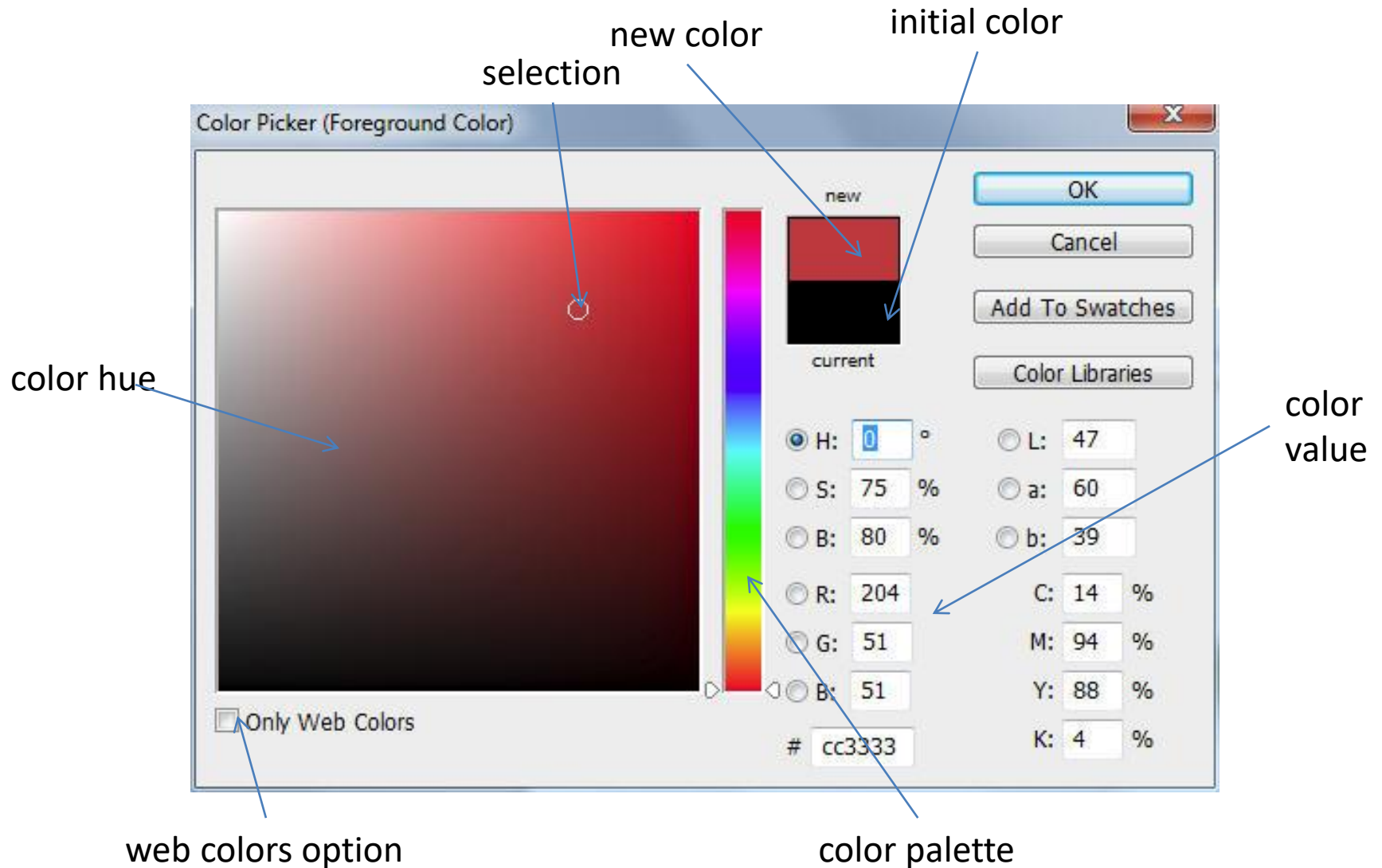
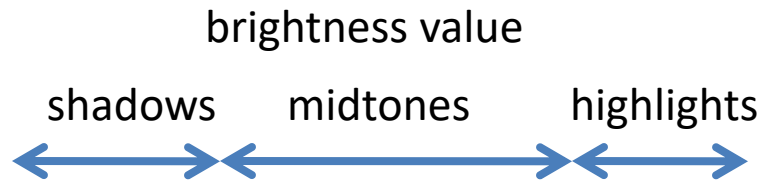
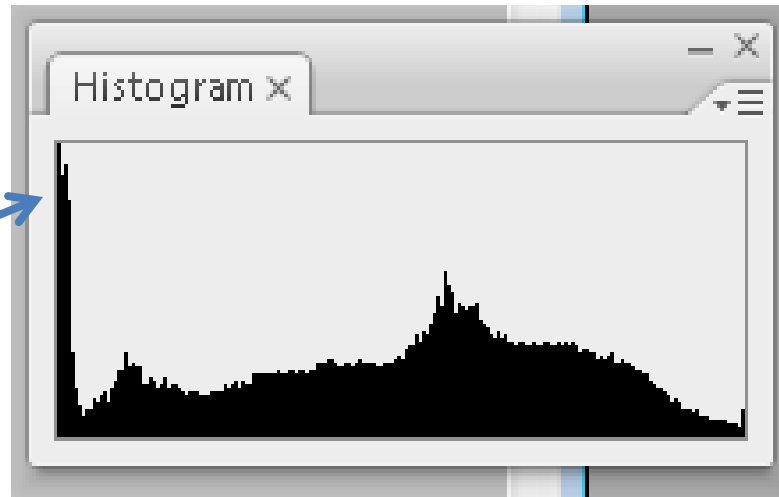


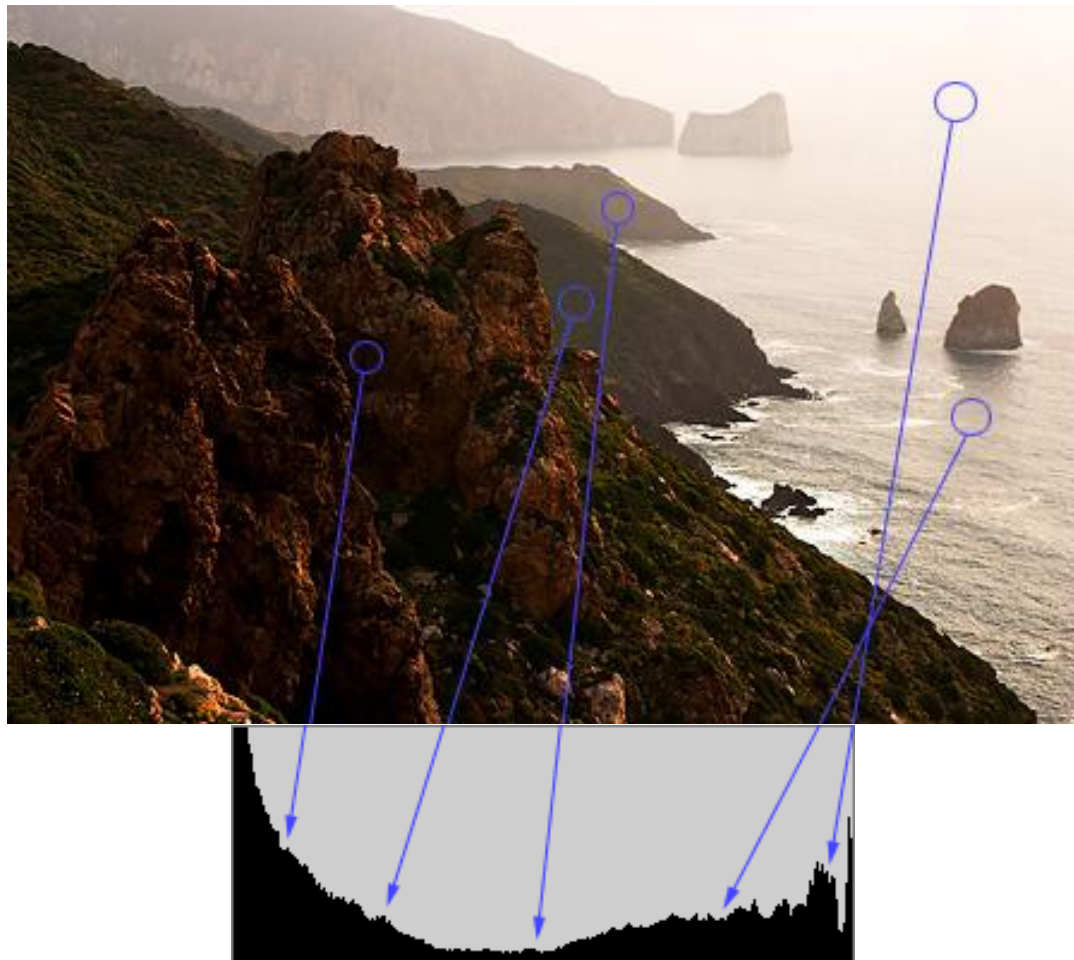
Image histogram



pixel number



Image



Image

Types of images

- grid of pixels (rasterized)
- vector format

Compression algorithms

- RLE
- Entropy Encoding

Image - Bitmap

Image Compression / Decompression

- **Algorithms:**

- with loss of information, lossy compression;
- without loss of information, lossless compression;
- symmetric (compression time = decompression time)
- asymmetric (the compression process takes longer than decompression)

Image - Bitmap

Algorithms	Data	Category
RLE, Huffman	Text, image	lossless
JPEG	Images	lossy
Fractal based	Images	lossy
DVI	Video	lossy
MPEG	Video	lossy

Image - Bitmap

Image Compression / Decompression

– *HOFFMAN*

- variable coding
- uses the frequency of symbols
- uses a binary unbalanced tree

– *RLE(Run Length Encoding)**

- good for data with low diversity and high repeatability;
- efficient when a small number of colors is used;

Image - Bitmap

Image Compression / Decompression

- *RGB555*

- reduces the total number of bits used for storing a color from 8 to 5;

- *JPEG*

- Hybrid compression algorithm using 2 techniques
 - cosine discrete transformation (TCD)
 - Huffman coding
 - a good compression rate of almost 15:1

- *MJPEG(Motion JPEG)*

- Animation standard;

Image

Bitmap format:

- is a simple matrix, each element representing the color of a pixel;
- the image has volume;
- the image quality is dependent on the visualization scale;
- the image cannot adapt to a variable visualization scale;
- as higher the compression rate is as lower the image quality is;

Image

Bitmap format:

- Digital equipment used for processing bitmap images:
 - Scanner (DPI – Dots Per Inch)
 - Movie scanners
 - Digital camera (JPG, Raw)
 - Web camera

Image

Bitmap format:

- Bitmap file storing formats:
 - BMP
 - ICO (icons – 32 X 32)
 - TIFF (Tag Image File Format)
 - DIB (Device Independent Bitmap)
 - DDB (Device Dependent Bitmap)
 - JPG (Joint Photographic Experts Group, a bitmap compressed format with JPEG algorithm)
 - GIF (a reduced file format used for transferring bitmap images)
 - PNG (Portable Network Graphics)

http://en.wikipedia.org/wiki/Comparison_of_graphics_file_formats

Bitmap

- A bitmap is an array of bits that specify the color of each pixel in a rectangular array of pixels.
- The number of bits devoted to an individual pixel determines the number of colors that can be assigned to that pixel.
- If each pixel is represented by 4 bits, then a given pixel can be assigned one of 16 different colors ($2^4 = 16$).

Internal structure of a BMP file

- **BITMAPHEADER**, 14 bytes, includes various data about the header of the file, such as:
 - **2 bytes**, the file signature, BM with value *4D42h*;
 - **4 bytes**, the file size;
 - **4 bytes**, reserved area;
 - **4 bytes**, the offset at which actual data begin;

Internal structure of a BMP file

- **BITMAPINFOHEADER**, fixed length, 40 bytes:
 - **4 bytes**, the size of the info header region;
 - Image height (**4 bytes**) and image width (**4 bytes**);
 - **2 bytes**, number of plans;
 - **2 bytes**, color depth, value given in number of bits per pixel;
 - **4 bytes**, compression, if it's the case;
 - **4 bytes**, the total size of the image, if compression is applied;
 - **4 bytes**, horizontal resolution and **4 bytes**, vertical resolution;
 - **4 bytes**, the number of used colors and **4 bytes**, the number of important colors.

Bitmap file structure

OPTIONAL PALLETE is used only if the bits per pixel value is lower or equal to 8;

IMAGE DATA is the area which contains each pixel value;

Image - Bitmap

- stores digital images;
- **BMP** (Microsoft **Windows Bitmap**)
- is also known as DIB (device - independent bitmap)
- extensions: .bmp or .dib
- the content is uncompressed, being very large;
- the size of the file is directly dependent on the size of the image (height and width) but also on the depth of the image (bit depth or color depth)

Color depth or bit depth:

- The number of bits used for pixel representation:
- Possible values equal to 1, 4, 8, 24 or 32 bits.
 - black and white images - 1 bit
 - 16 colors images - 4 bits
 - 256 colors images - 8 bits
 - images with $16,7 * 10^6$ colors (24 bits, one byte for each of the three colors in RGB representation)
 - images with transparency (RGB model + 1 byte for transparency) a total of 32 bits

Image - Bitmap

Offset (h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00000000	42 ¹	4D	46	00 ²	00	00	00	00 ³	00	00	36	00 ⁴	00	00	28	00
00000010	00	00	02	00	00	00	02	00	00	00	01	00	18	00	00	00
00000020	00	00	10	00	00	00	00	00	00	00	00	00	00	00	00	00
00000030	00	00	00	00	00	00	00	00 ⁵	FF	00	FF ⁶	00	00 ⁷	00	FF	00 ⁸
00000040	00	FF	FF ⁹	FF	00 ¹⁰	00										

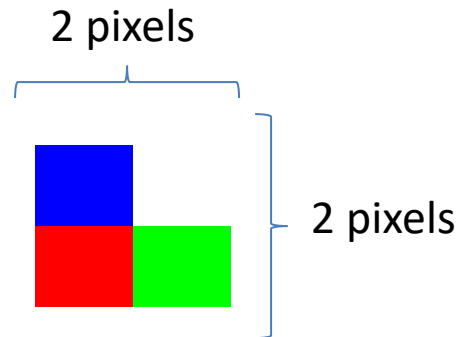


Image - Bitmap

Structure		Legend
Bitmap Header		14 Bytes area
	1	2B , file signature, <i>4D42h</i>
	2	4B , file size, <i>00000046h</i>
	3	4B , reserved area, <i>00000000h</i>
	4	4B , image data offset, <i>00000036h</i>
Bitmap InfoHeader		40 Bytes area
Bitmap Data (pixels)		Pixels stored in BGR format / 1B per color
	5	3B pixel – Red color, <i>0000FFh</i>
	6	3B pixel – Green color, <i>00FF00h</i>
	7	2B padding after full line
	8	3B pixel – Blue color, <i>FF0000h</i>
	9	3B pixel – White color, <i>FFFFFFh</i>
	10	2B padding after full line

User defined types

The article is a collection of heterogeneous data stored in a compact region of memory. It allows users to bring together multiple variables in a single defined type entity.

```
struct name {  
    type    member_name;  
};
```

User defined types

- Bits structures allow bit access level;
- To access data at bit level programmers must:
 - define a structure which maps perfectly on the data which must be accessed;
 - load the pointer to the bit structure with the address of the respective data;
 - processing the bits without altering the meaning and internal representation of data;

User defined types

- Bit fields:

```
struct name {  
    type field_name      : width;  
    int                   : width;  
    ...  
    int    field_name    : width;  
};
```

User defined types

- The declared bit field has the following properties:
 - The type may be int, signed or unsigned;
 - The number of bits that a member can refer is specified by width;
 - If member name is missing, the number of bits specified in the width field can not be referred and accessed;
 - The fields are accessed the same way as in any other structure;
 - The address of fields can't be used;

Bitwise operators

&

- AND

|

- OR

^

- OR exclusive - XOR

~

- One complement (flips bit value)

>>

- Right shifting

<<

- Left shifting

Bitwise operators

unsigned char x	x after operation	value of x
x=7	0000 0111	7
x=x<<1	0 0000 111 0	14
x=x<<3	000 0111 000 0	112
x=x<<2	01 1100 000 0	-64
x=x>>1	1 110 0000 0	-32
x=x>>2	11 11 1000 00	-8

Bitwise operators

& (AND)	0	1
0	0	0
1	0	1

 (OR)	0	1
0	0	1
1	1	1

^ (XOR)	0	1
0	0	1
1	1	0

Bitwise operators

11000001
01111111
& 01000001

AND

10000000
00000011
| 10000011

OR

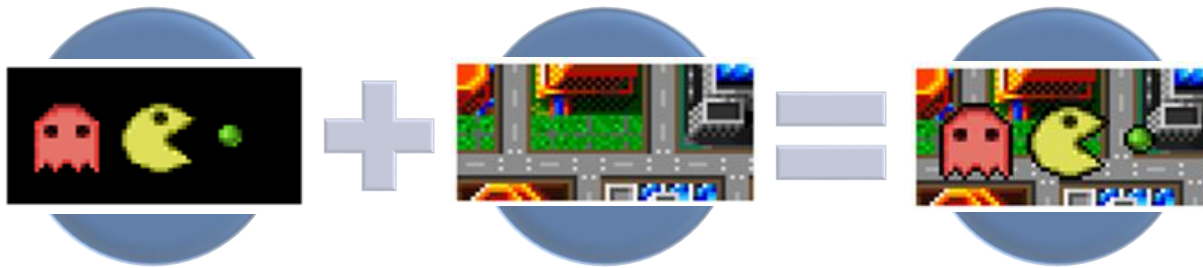
01111111
01111000
^ 00000111

XOR

00101100
~ 11010011
~ 00101100

NOT

Bitwise operators



Bitwise operators



Image

Big picture example*

- President Barack Obama's Inaugural Address by David Bergman
 - elements: 220 images
 - dimension: 59,783 X 24,658 pixels
 - size: 1.47 gigapixels
 - <http://gigapan.org/viewGigapan.php?id=15374>

Image - Bitmap

VCX (PC PaintBrush File Format):

- known as PaintBrush on Windows platforms;
- stores a 8 bit color image (256 colors),
maximum dimension of 64000 * 64000 pixels
- the compression algorithm used is RLE

Image - Bitmap

TIFF (Tag Image File Format)

- Known for storing and transferring of scanned images;
- Uses different compression algorithms and it's very efficient: RLE, LZW (Lempel-Ziv-Welch) or JPEG.
- It platform independent => high level of portability

Image - Bitmap

ICO (Icon Resource File)

- It is a *bitmap* format
- Stores small size images, used in Windows for representing pictograms
- Uses different resolutions and color pallets;
- It is easily obtained from normal images (<http://www.favicon.co.uk/>)

Image - Bitmap

JPG (Joint Photographics Experts Group)

- Stores *bitmap* images in a compressed format
- Has a very good compression rate
- Is defined as being a standard in image compression and an image file format;

Image - Bitmap

GIF (Graphics Interchange Format)

- is mostly used as a web based format;
- used for storing small size bitmap images;
- has high compression rates;
- developed by CompuServe for delivering image content easier on the web;
- uses the LZW compression algorithm.

Image - Bitmap

PNG (Portable Network Graphics)

- It is an open format;
- Uses compression without loss of information;
- Used in web applications for image transfer;
- Created for replacing GIF format;

Image – Vector format

- It deals with picture semantic (drawings → points and mathematical functions)
- Images have small sizes;
- Images are view scale independent;
- It can replace bitmap because not everything can resume to points and functions; eg: landscapes;
- The color and pixel position is determined by mathematical functions;

Image – Vector format

- DXF (Drawing Exchange Format – for Autocad made by Autodesk)
- EPS (Encapsulated PostScript – Adobe format și lb. PostScript)
- CGM (Computer Graphic Metafile) for transferring images across platforms;
- SVG (Simple Vector Graphics) defined for vector based web graphics

Steganography Overview



Security through
obscurity



Steganography
historical facts



Steganography
overview



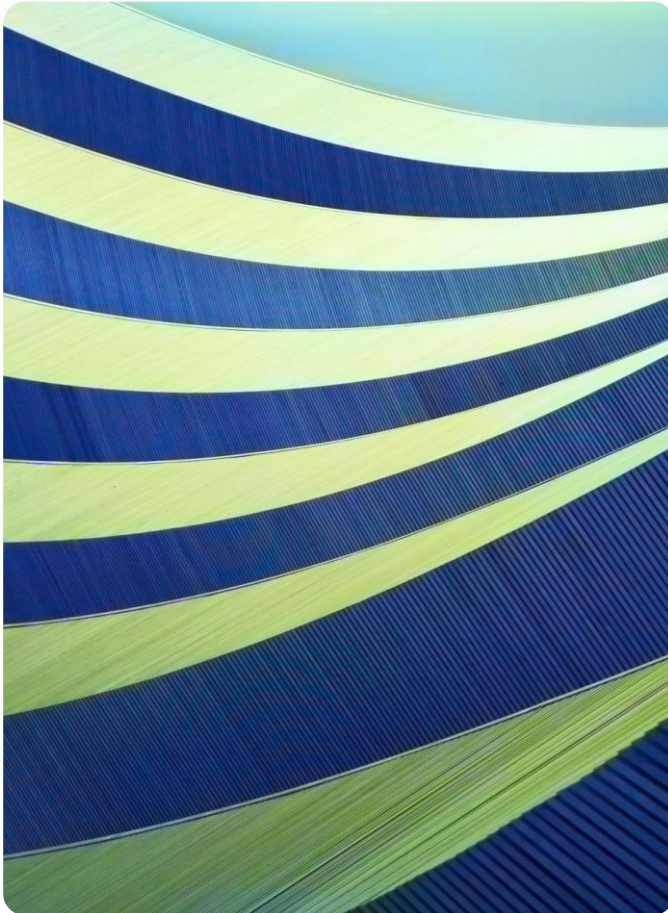
Doing
steganography



Undoing
steganography



Steganalysis
techniques



Steganography

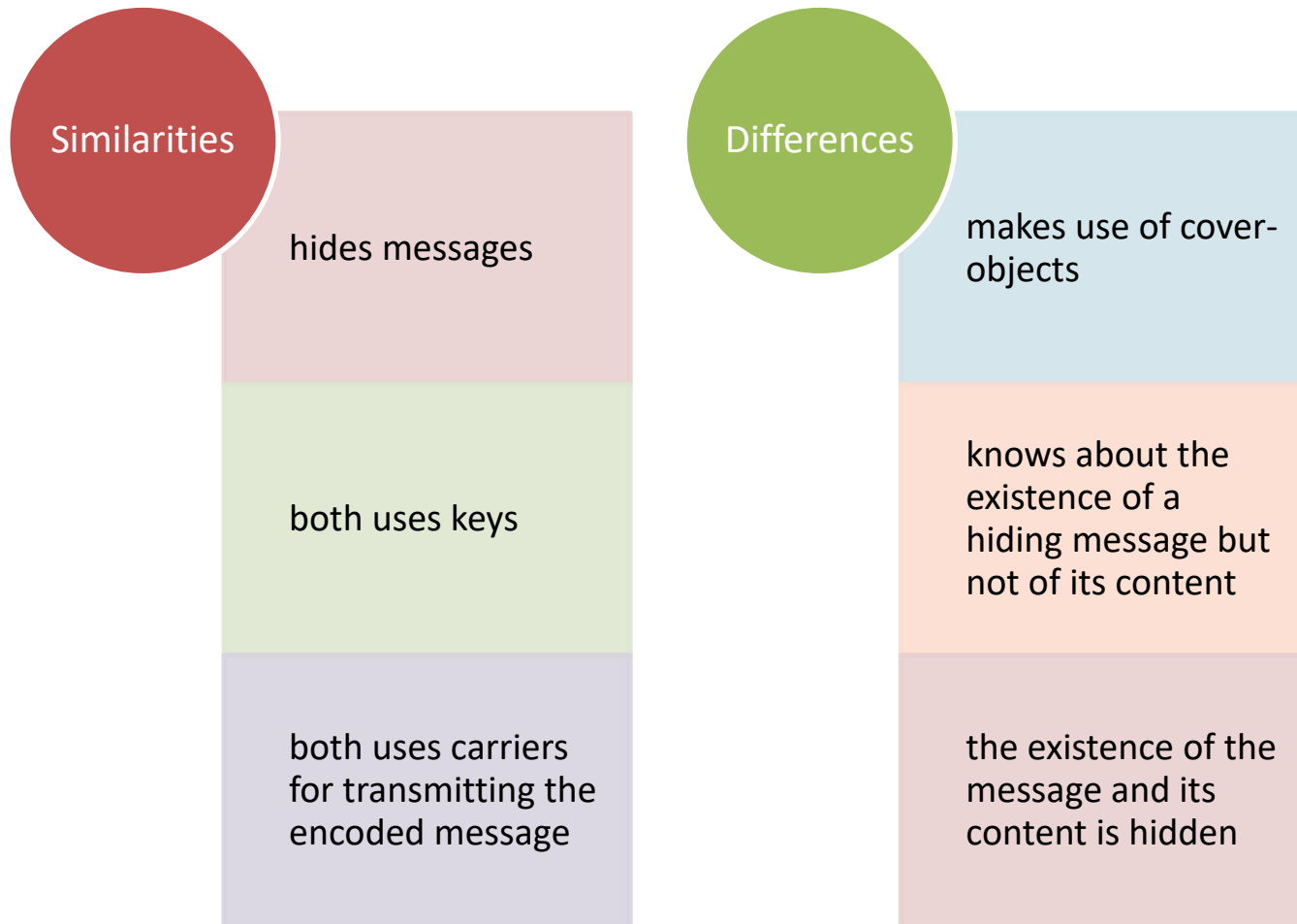
Steganography is the science of hiding the existence of messages embedded and passed from source to recipient into other files.

Steganography = steganos + graph = hidden writing

steganos = hidden

grapho = writing

Steganography vs. Cryptography



Steganography principles

- Pure steganography
 - The principle used is similar to what “*Security through obscurity*” principle states, which is: the methods used in the encoding processes should not be public knowledge or hide anything about the encoding process.
- Key based steganography
 - uses public or secret key
 - Auguste Kerckhoffs’s principle : “*A cryptosystem should be secure even if everything about the system, except the key, is public knowledge.*”
- NIST security principle:
 - “*System security should not depend on the secrecy of the implementation or its components.*” is a more recent version of Kerckhoffs’s doctrine.

Main objective

Steganography
conceals information
in other, seemingly
innocent media.



Tackled security issues

- transmitting information in a secure manner;
- uses cryptography for hiding the private key;
- protect embedded message from tampering;
- conceals the existence of the message inside the cover object;
- prevents the extraction of the hidden message;

Types of Steganography

Cryptographic aspects:

Pure steganography

Key-based steganography

Cover-object manifestations:

Digital steganography

Physical steganography

Types of cover-object:

WLAN packages

Steganophony

Images

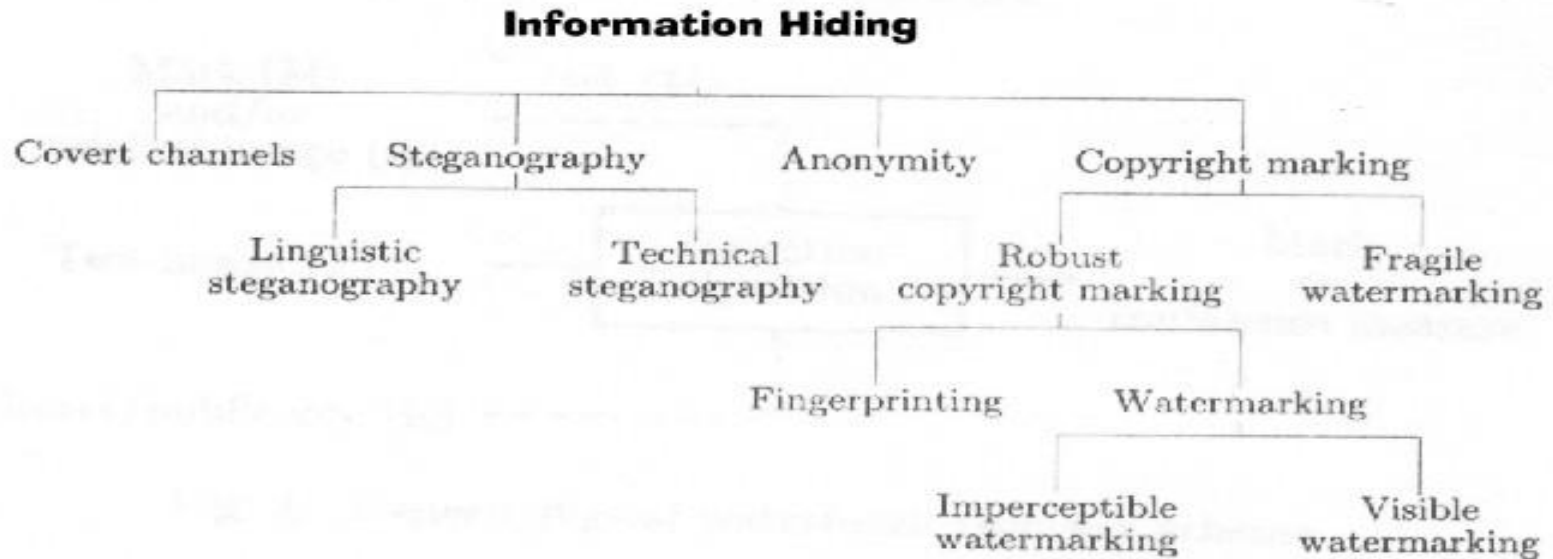
Sounds

Video

Steganography historical facts

- Herodotus (484-425 BC) – security through obscurity
- Pliny the Elder (23 – 79 AD) – first invisible ink
- Renaissance Steganography
 - (1499 AD), Johannes Trithemius – first printed book on cryptology
 - (1665 AD), Steganographica, Gaspari Schotti
 - Giovanni Battista Porta – invisible ink on a hard boiled egg shell
 - 1870 – The Pigeon Post into Paris;
- Modern age
 - 1945 – the use of invisible ink in the second world war
 - 2001 – film industry, A beautiful mind

Steganography



A. P. Petitcolas, R. J. Anderson, M. G. Kuhn,
“Information Hiding – A Survey”, Proceedings of the
IEEE, special issue on protection of multimedia
content, 87(7):1062-1078, July 1999

Steganography vs. Watermarking

Steganography: techniques specific to a 1 to 1 communication;

Watermarking: techniques specific to a 1 to many communication;

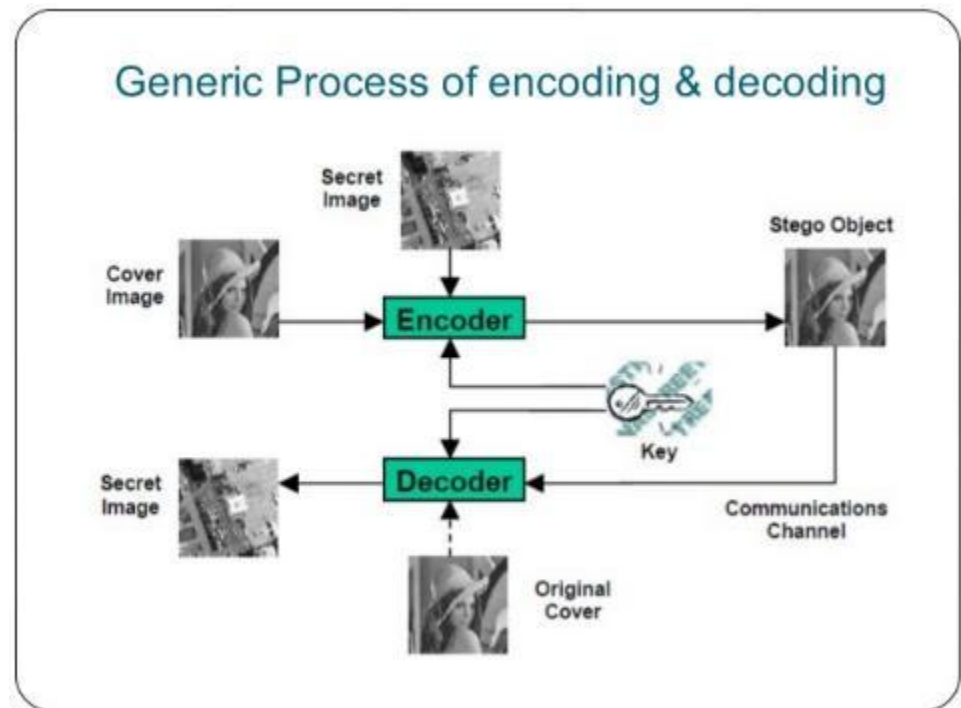
Steganography elements

Involved components:

- Embedded message
- Cover-object
- Stega-key
- Encoding algorithm
- Stego object

Characteristics:

- Total capacity
- Robustness
- Undetectable
- Security



Steganography

Cover file is the document in which data is hidden;

Embedded file is the document hidden into the cover file.

Steganography

- Main objective:
 - Hiding data in other data;
- Secondary objectives:
 - Prevent the discovery of a message into a cover object
 - Prevent the extraction of an embedded message (robustness)
 - Prevent that the message will suffer as little damage as possible when altering the carrier

Embedded message

- Must be of optimal length
- Must not include something related with the cover-file
- Must know and analyze the cover object in order to determine what type of carrier can be used
- Can be additionally encrypted to offer a better protection in case the message is found

Cover file

Images:

- alteration of the least significant bit;
- applying masks and filters;
- image transformation algorithms;

Sound:

- encoding with the use of the LSB by substitution;
- encoding a message by using the parity bit;
- phase coding using the Discrete Fourier Transformation;
- spread spectrum using the frequency spectrum;

Video:

- conventional, image oriented; intra frame, spatial domain, transform domain;
- using compression techniques combined with steganography by using movement vectors or error prediction rate for hiding messages;
- spatial and temporal prediction;

Cover transformations

- Spatial domain techniques
 - Least Significant Bit (LSB)
- Frequency domain techniques
 - Discrete Cosine Transform (DCT)
 - Low frequencies (important information)
 - High frequencies (edges of the image)

Steganography keys

- Can be private keys with the use of an symmetric encryption algorithms, such as: AES based on Rijndael cipher;
- Public key systems can also be used using asymmetric encryption algorithms, such as: RSA based on two large prime numbers;
- The keys must be of optimal length (256 for AES, 2048 for RSA);

Steganography approaches

- Statistical formulation:
 - False Positive $P_{fp} = P(\text{detect message} \mid \text{no message})$;
 - Detection Probability = $P(\text{detect message} \mid \text{message})$;
 - Dispersion and Distribution;
- Mathematical algorithms:
 - Discrete Cosine Transformation;
 - Discrete Fourier Transformation;
- Information theory means:
 - Embedding algorithms;
 - Cover-object analysis;

Steganography Algorithms

Substitution systems: Least significant

Bit Substitution – very weak

Domain transformation: Discrete

Cosine Transform – weak

Spread spectrum techniques – Direct-

Sequence or Frequency-Hopping

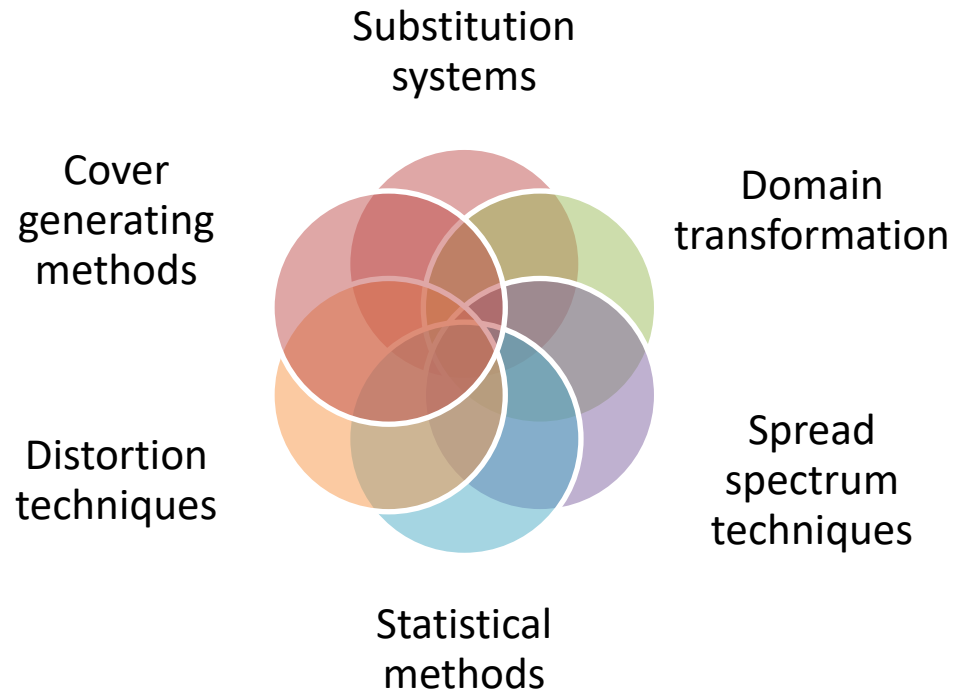
Scheme – good

Distortion techniques – need access to

the original cover – major flaw

Cover generating methods – resource

consuming



Steganography – Steganalysis

Steganalysis is the science which studies the techniques used for concealing messages inside cover files.

Steganalysis identifies the presence of an embedded message into a cover file.

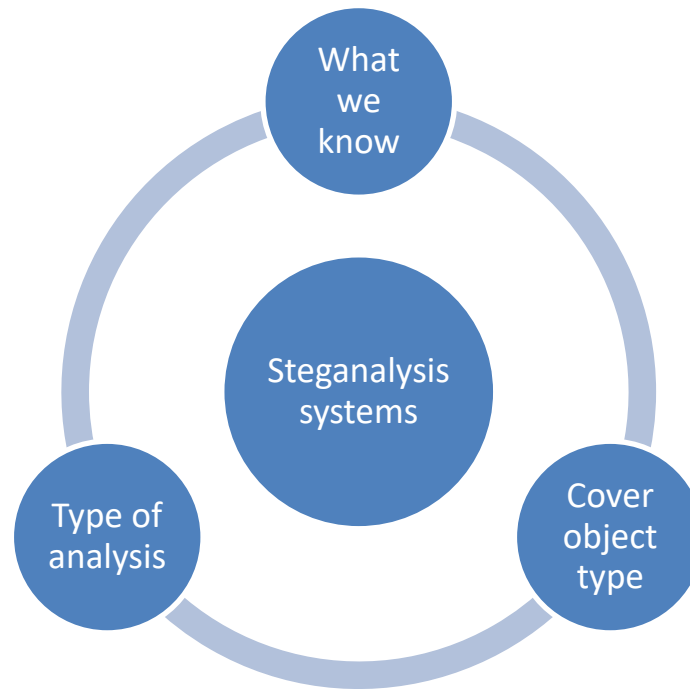
Steganalysis

The objectives are to accurately identify the presence of an embedded message, keep the integrity of the message when trying to retrieve it and minimize the false positive alerts.

Steganalysis concept

- Steganalysis reflects upon the steganography techniques applied in cover objects
- Identifies if a possible cover object contains or not an embedded message
- The art or science of discrimination between hidden message and cover object
- Must identify and analysis noise in the cover-objects
- Nonetheless message does not have to be extracted

Steganalysis classification systems



The available entropy about stega-content

Only the stega-object is available

The stega-
algorithm is known

The original cover
object is available

Steganalysis techniques cover type based

- Analysis of alteration of the least significant bit;
- Identifying mask and filters applied to spatial domain;
- Analysis of Domain Transformation;
- Statistical analysis of the parity bit;
- Frequency spectrum analysis;
- Intra/Inter frame structural analysis;
- Spatial and temporal prediction analysis;

Types of analysis

- Statistical steganalysis
 - Least Significant Bit
 - Spread Spectrum Analysis
 - Discrete Cosine Transform
- Target based steganalysis
 - The destruction of the message
 - The analysis of the cover object
 - The identification of message presence

Steganalysis

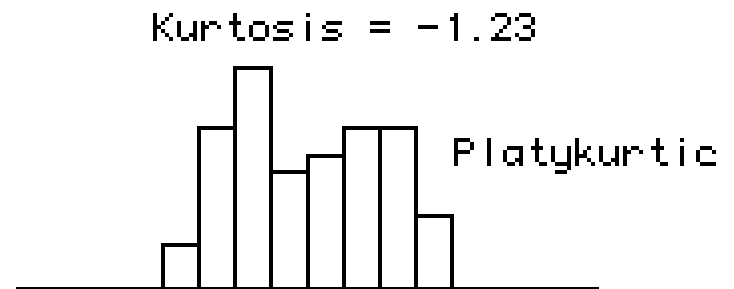
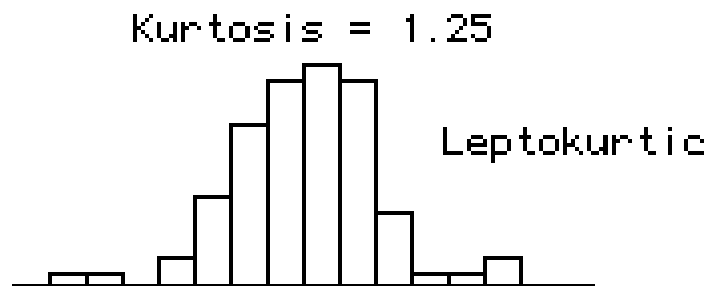
- Methods of detecting the use of Steganography
 - Visual Detection (JPEG, BMP, GIF, etc.)
 - Audible Detection (WAV, MPEG, etc.)
 - Statistical Detection (changes in patterns of the pixels or LSB – Least Significant Bit) or Histogram Analysis
 - Structural Detection - View file properties/contents
 - size difference
 - date/time difference
 - contents – modifications
 - checksum

Methods of detection

- Categories
 - Anomaly
 - Histogram analysis
 - Change in file properties
 - Statistical Attack
 - Visually
 - Audible
 - Signature
 - A pattern consistent with the program used

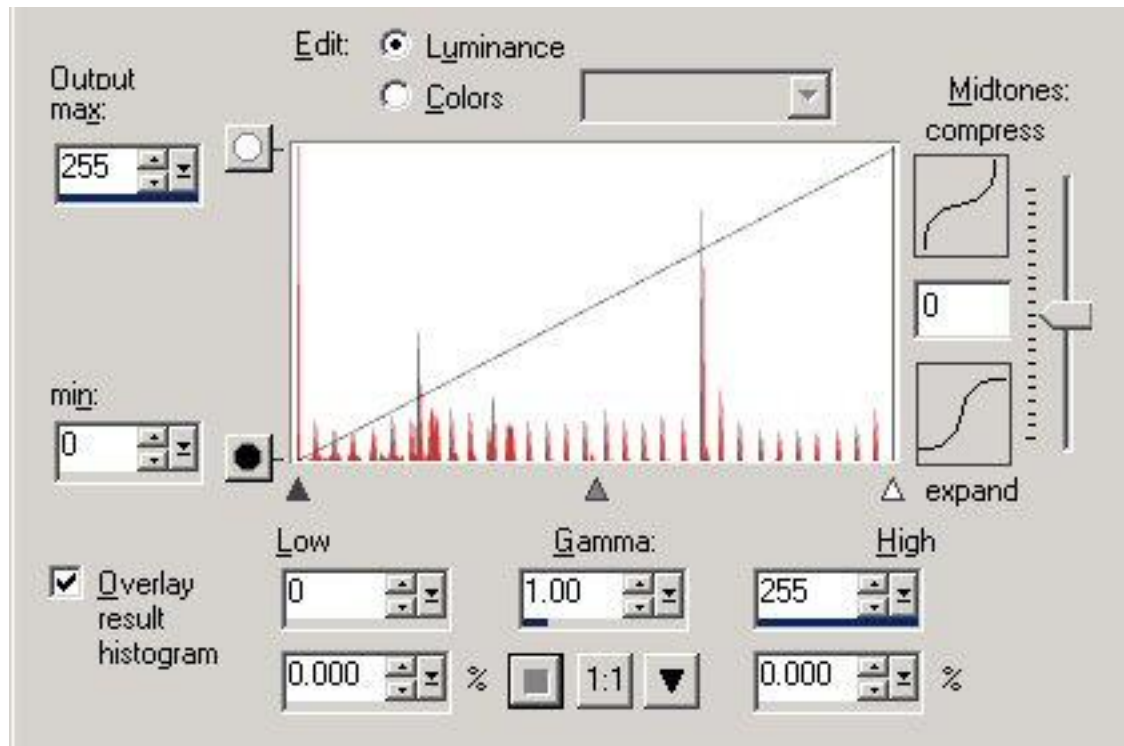
Methods of detection

- Kurtosis
 - The degree of flatness or peakedness of a curve describing a frequency of distribution
 - Random House Dictionary



Methods of detection

- By comparing histograms, we can see this histogram has a very noticeable repetitive trend.



Steganography



Steganography

1. Can use a key for altering the original message, this way preventing message decryption if the hidden content was revealed;
2. Must use a function for spreading the message through the entire content of the cover file;
3. Combining the key with the dispersion function for decreasing the detection degree;

Steganography

Steganography image techniques:

- Modifying the least significant bit;
- Applying masks and filters;
- Image transformation algorithms;

Steganography

Hiding a character at the least significant bit of each color

24 bit bitmap => 3 bytes / color

Value	Cod	Pixel	Hidden character
10101111	R	1px.	A = ASCII (65 Dec / 41 Hex)
01101110	G		
11111110	B		
10101110	R	2px.	
01101100	G		
11111110	B		
10101111	R	3px.	
01101101	G		
11111111	B		

Integrity Verification

	$\mathbb{R}^2$ IDMEAL	$\mathbb{R}^2$ DESCRIPTION	$\mathbb{R}^2 \uparrow^2$ NOKCAL	$\mathbb{R}^2$ NOPROTEINS	$\mathbb{R}^2$ NOLIPIDS	$\mathbb{R}^2$ NOCARBO	$\mathbb{R}^2$ NOFIBERS	$\mathbb{R}^2 \uparrow^1$ TYPE	IMAGE
1	123 1123		107,25	9,75	7,5	0,75	0	1 (BLOB)	
2	67 167		347,5	123,2	19,265	188,75	26,45	1 (BLOB)	
3	64 164		403,75	127,5	7,725	505,5	91,5	1 (BLOB)	

STEG_BLOB = STEG (BLOB, HASH(67 || 167 || 347,5 || ... || 1))

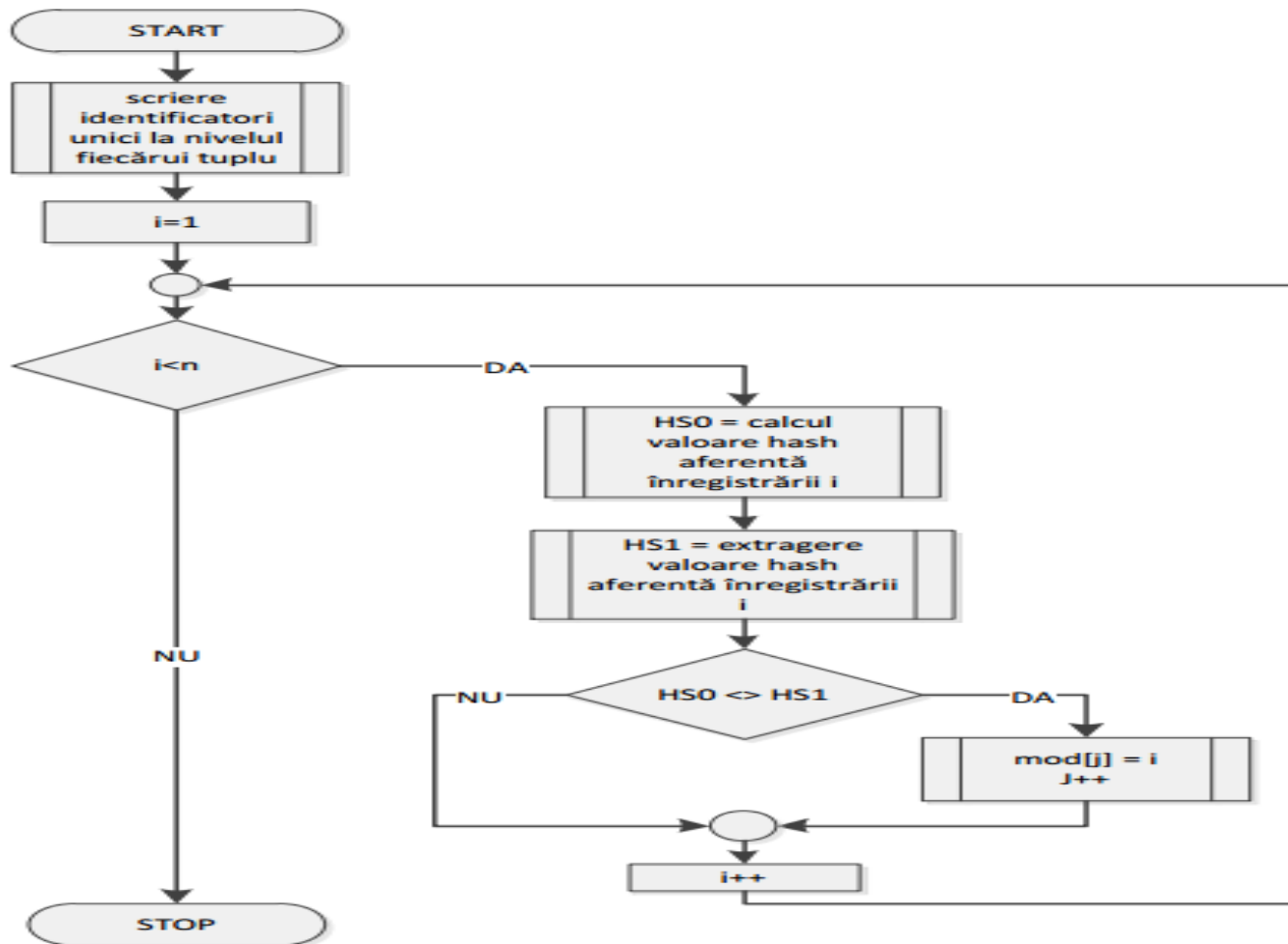
BEFORE



AFTER



Integrity Verification



Steganography - tools

- Steganos
- S-Tools
- StegHide
- Invisible Secrets
- JPHide
- Camouflage
- Hiderman

Is what you expected?

This was [Section 2.2 – Imagine](#)

Color pallets (RGB, CMYK, HSB, Lab, Grayscale, Web-safe)

Characteristics (Hue, Brightness, Saturation, Contrast,
Color balance, Bit depth, Resolution)

Types of images (bitmap, vector)

File formats (TIFF, ICO, JPG, GIF, BMP, PNG, EPS, SVG)

Compression

Steganography

Watermarking

Bibliography for **Section 2.2 – Images**

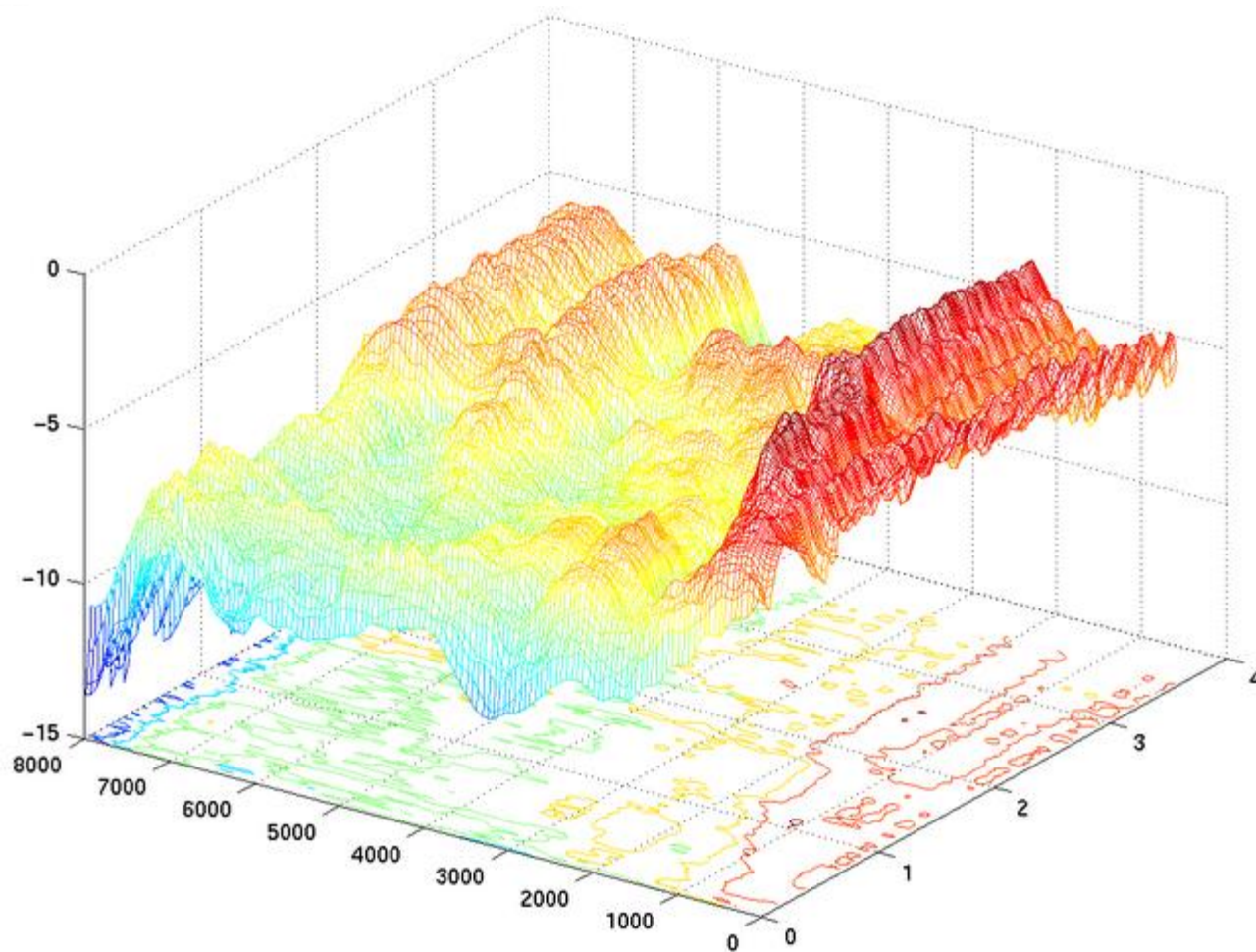
Auguste Kerckhoffs, “La cryptographie militaire”, Journal des sciences militaires, vol. IX, pp. 5–38, Jan. 1883, pp. 161–191, Feb. 1883.

Karen Scarfone, Wayne Jansen and Miles Tracy, “Guide to General Server Security”, National Institute of Standards and Technology, 2008

P. Petitcolas, R. J. Anderson, M. G. Kuhn, “Information Hiding – A Survey”, Proceedings of the IEEE, special issue on protection of multimedia content, 87(7):1062-1078, July 1999

Zaidoon Al-Ani, A.A. Zaidan, B.B. Zaidan and Hamdan Alanazi, “Overview: Main Fundamentals for Steganography”, Journal of Computing, vol. 2, Issue 3, March 2010, ISSN 2151 - 9617

2.3 Sound



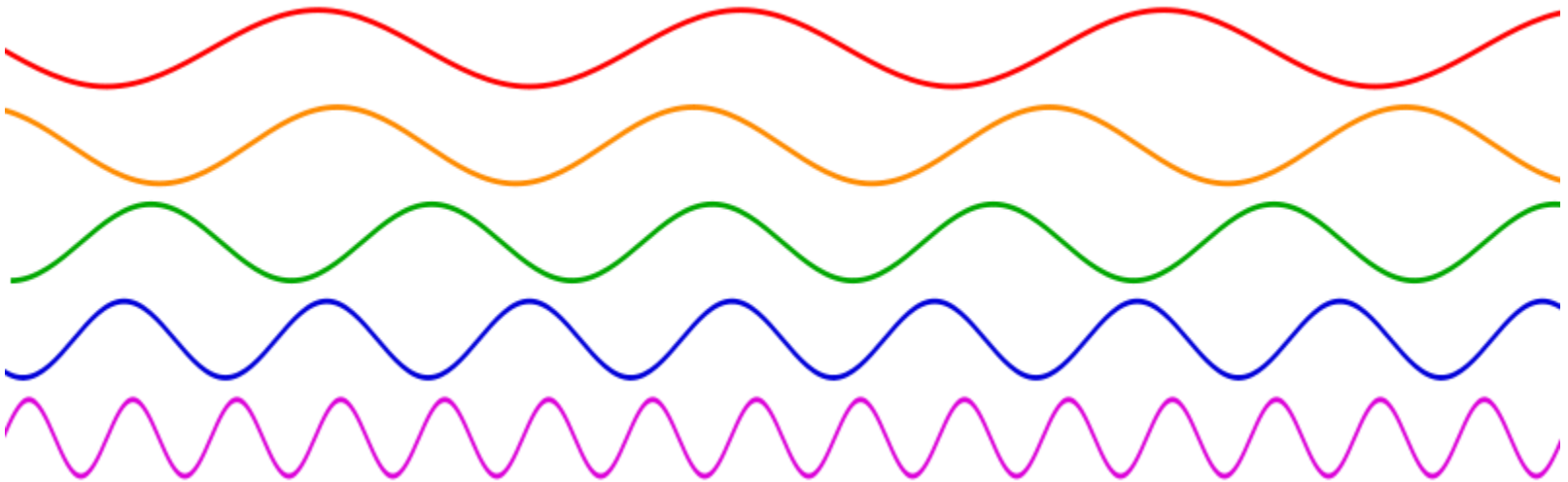
Audio - Sound

Characteristics:

- Frequency
- Wavelength
- Amplitude
- Intensity
- Speed
- Direction
- Pitch

Audio – Analogous Sound

Frequency



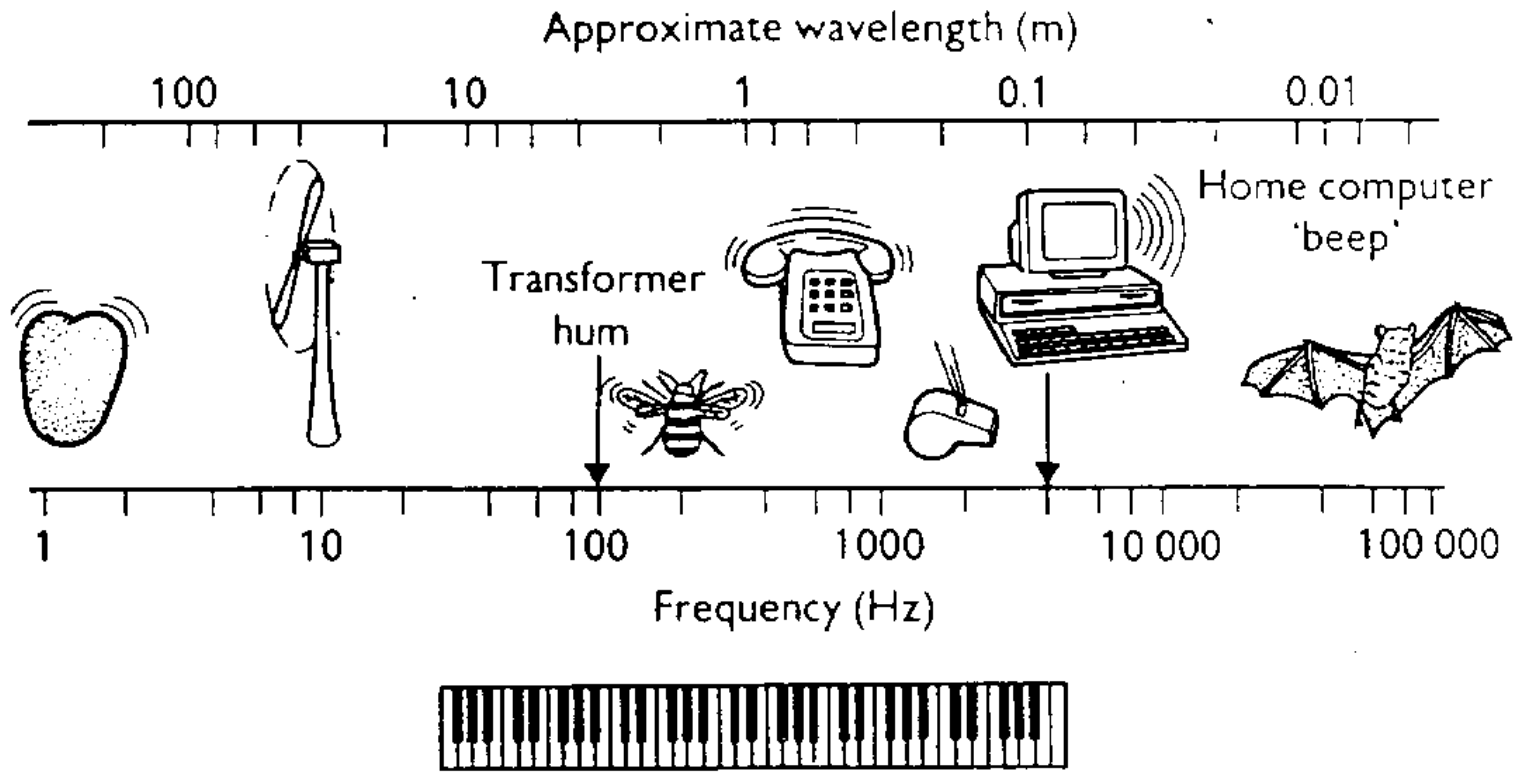
[WIKI]

Audio – Analogous Sound

Frequency:

- is measured in Hz (Hertz)
- 1 Hz -> one oscillation per second
- We can perceive frequencies between: 20 - 20,000 Hz

Audio – Analogous Sound



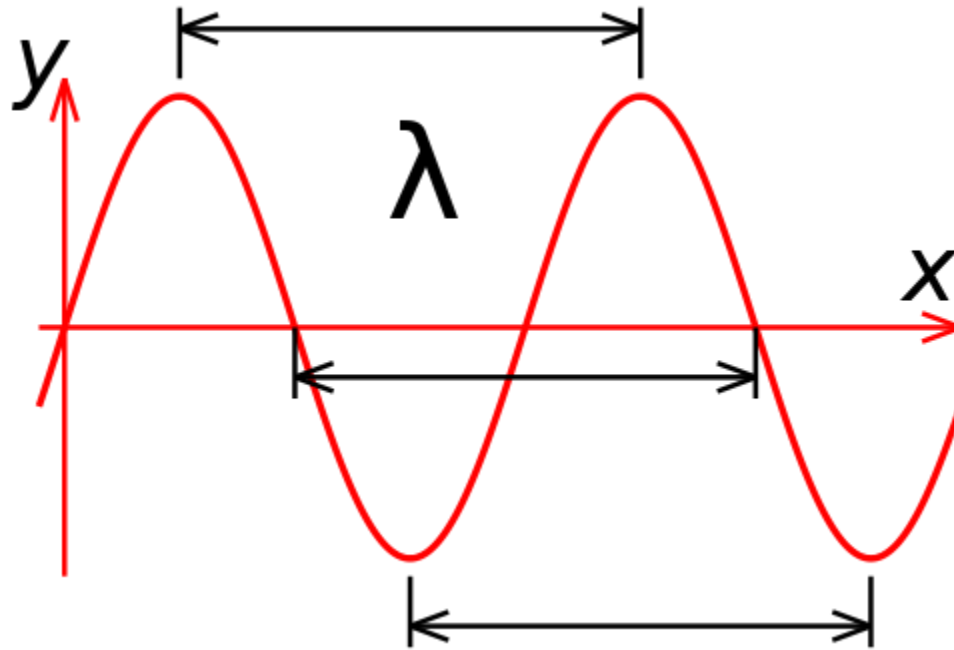
Pitch

pitch = frequency of sound

- middle C in equal temperament = 261.6 Hz
- the **pitch** of a **sound** is determined by the rate of vibration, or frequency, of the **sound** wave

Audio – Analogous Sound

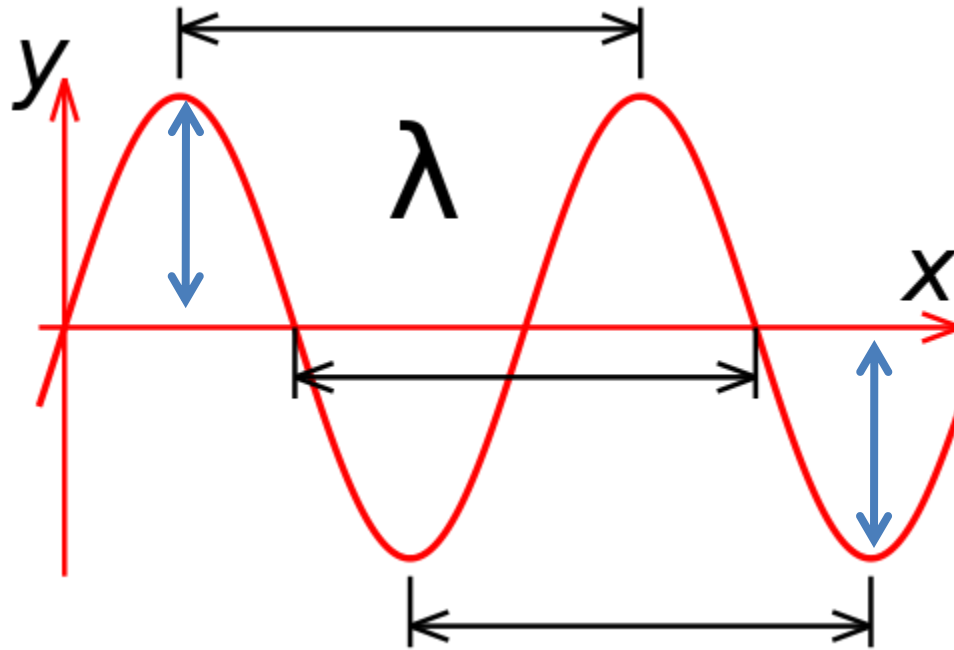
Wave length



[WIKI]

Audio – Analogous Sound

Amplitude



[WIKI]

Intensity

The sound intensity may be expressed in decibels above the standard threshold of hearing I_0 .

$$I(dB) = 10 \log_{10} \left[\frac{I}{I_0} \right]$$

$$I(dB) = 10 \log_{10} \left[\frac{10,000 I_0}{I_0} \right] = 10 \cdot 4 dB = 40 dB$$

Speed

The speed of sound is the distance travelled per unit time by a wave sound as it propagates through an elastic medium.

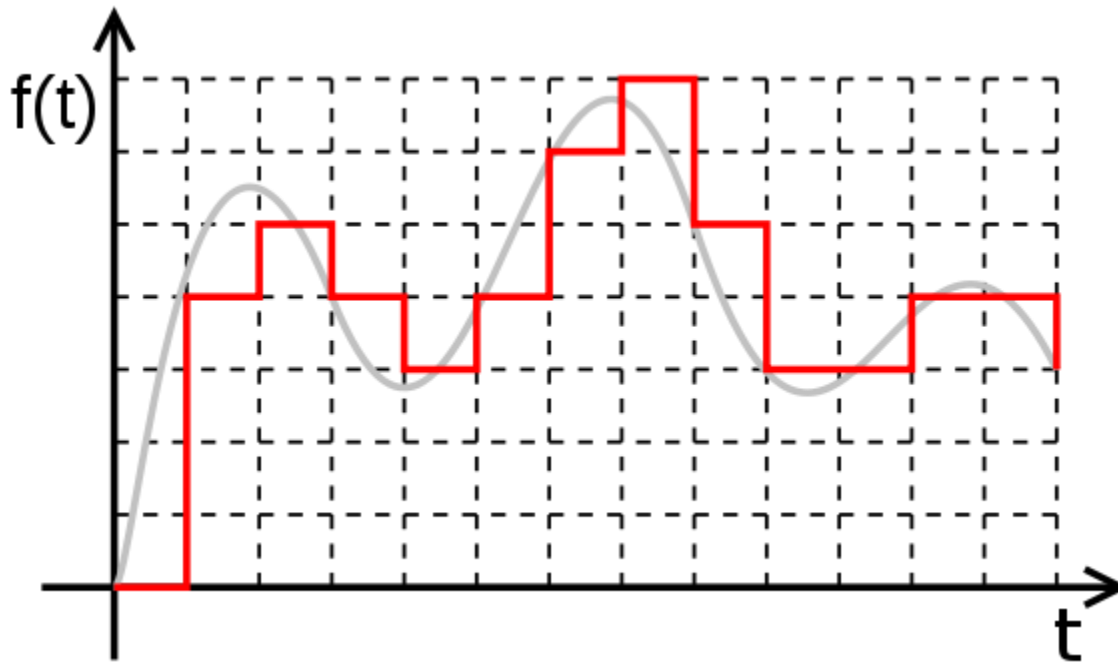
The speed of sound in dry air is approx.:

$$(331.4 + 0.6 * T_c) \text{ m/s}$$

Audio – Analogous Sound

Medium	Speed (m/s)
Air	343
Water	1,372
Concrete	3,048
Glass	3,658
Iron	5,182
Lead	1,219
Steel	5,182
Hard Wood	4,267
Soft Wood	3,353

Audio – Digital Sound



[WIKI]

Audio - Codec

Used for encoding and decoding sound



It also uses sound compression algorithms

with loss of information,
lossy compression

without loss of information,
lossless compression



An audio file often uses a specific type of
codec



Audio - Codec

MP3

- MPEG-1 or MPEG-2 Audio Layer 3
- Defined in 1993 and widely used over the Internet
- Uses a compression algorithm with loss of information (lossy compression): at 128 kbit/s the mp3 audio file is almost 11 times smaller than its audio cd equivalent (.cda)
- The compression is based on eliminating some frequencies that are not perceived by the human ear

Audio - Codec

MP3

- The compression is made by setting a *bit rate* which gives the number of Kb used for each audio second
- The audio quality is directly dependent on the size of the audio file
- Possible bit rate values: 32, 40, 48, 56, 64, 80, 96, 112, 128, 160, 192, 224, 256 and 320 kbit/s
- For a CD audio track, the bit rate is of 1,411.2 kbit/s

Audio - Codec

MP3 bit rate:

- *variable bitrate* (VBR)
- *constant bitrate* (CBR)
- 1997 the first audio player – Winamp
- 1998 the first portable MP3 player – MPMan

Audio - Codec

FLAC

- Free Lossless Audio Codec
- Launched in 2001; a more stable version was hardly released in 2007
- Implements a lossless compression algorithm
 - the quality is identical with the raw file
- The compression rate is almost 2:1

Audio Formats

- Ways of storing sound in digital format
- Audio formats classified by compression:
 - without compression: WAV, AIFF, .cda (Audio CD Track) (~ 10 MB per minute);
 - with a maximum of 2:1 lossless compression: FLAC, Apple Lossless, MPEG-4 SLS, MPEG-4 ALS, MPEG-4 DST, Windows Media Audio Lossless (WMA Lossless);
 - with lossy compression: MP3, Windows Media Audio (WMA);

Audio

Algorithm	Object	Category
ADPCM	Sound	Lossy compression
MPEG - Audio	Sound	Lossy compression

Uncompressed audio format

WAV based on RIFF (Resource Interchange File Format)

AIFF based on IFF (Interchange File Format)

44.100kHz, 16 bit, 2 channels =>

$(44100 * 16 * 2) = 176,4 \text{ KBytes/sec. (bitRate)}$

WAVE - PCM

Field	Length	Content
ckID	4	Chunk ID: "RIFF"
cksize	4	Chunk size: $4+n$
WAVEID	4	WAVE ID: "WAVE"
WAVE chunks	n	Wave chunks containing data about formats and actual content

WAVE - PCM

For a file with the following characteristics:

- N_c channels
- N_s the total number of blocks
- each block containing N_c samples
- sampling rate is F (blocks per second)
- each sample has a total of M bytes

Field		Length	Content
ckID		4	Chunk ID: "RIFF"
cksize		4	$4 + 24 + (8 + M * N_c * N_s + (0 \text{ or } 1))$
WAVEID		4	WAVE ID: "WAVE"
ckID		4	Chunk ID: "fmt "
cksize		4	Chunk size: 16
wFormatTag		2	WAVE_FORMAT_PCM
nChannels		2	N_c
nSamplesPerSec		4	F
nAvgBytesPerSec		4	$F * M * N_c$
nBlockAlign		2	$M * N_c$
wBitsPerSample		2	rounds up to $8 * M$
ckID		4	Chunk ID: "data"
cksize		4	Chunk size: $M * N_c * N_s$
sampled data		$M * N_c * N_s$	$N_c * N_s$ channel M -byte samples
pad		0 or 1	Padding byte if $M * N_c * N_s$ is odd

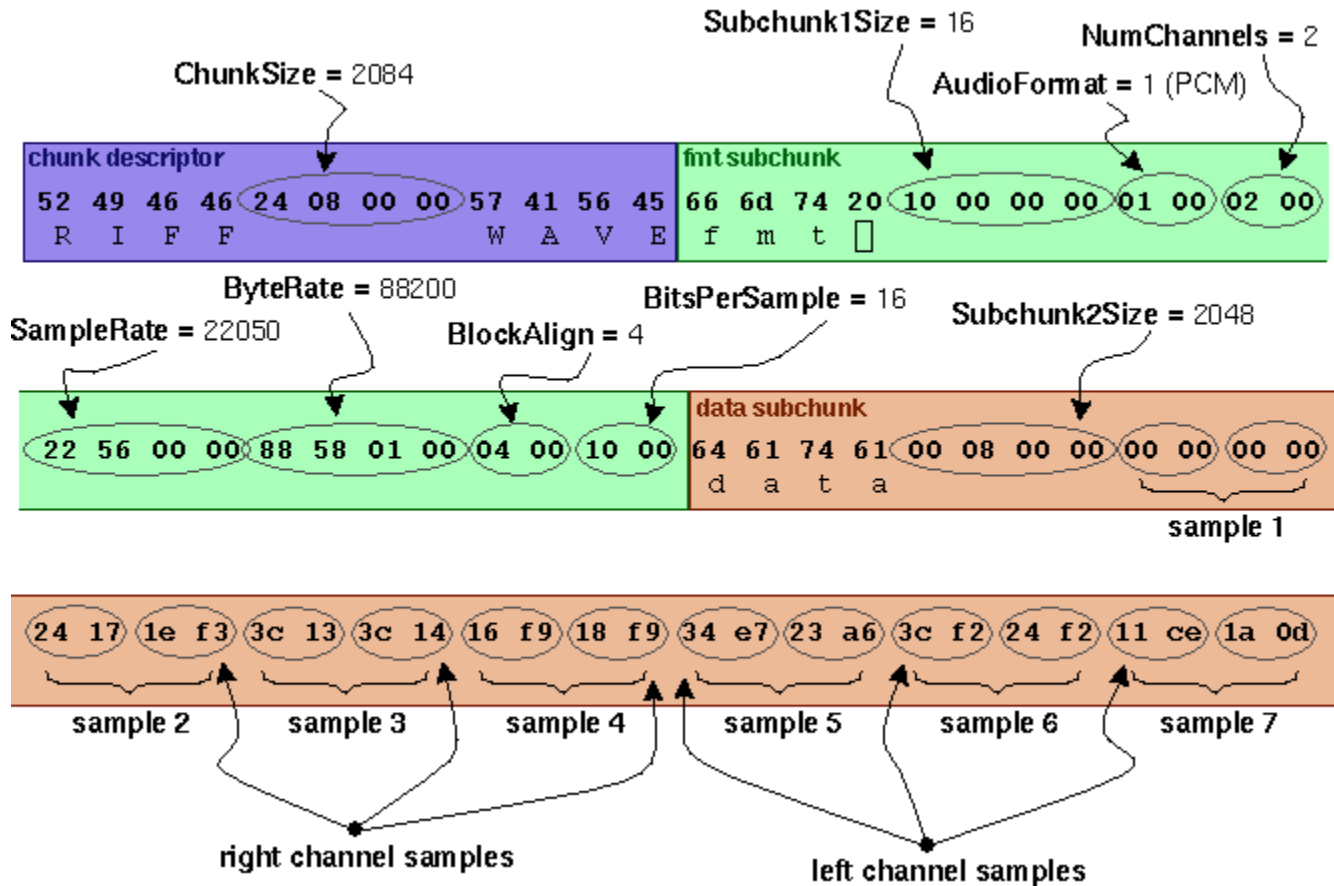
RIFF - Wave

Byte Offset	Name	Field Size(bytes)	Notes	
0	ID	4	Letters "RIFF"	RIFF Section
4	Size	4	Size of this file less 8	
8	Format	4	Letters "WAVE"	
12	ID	4	letters "fmt"	Format section
16	Size	4	Size of format section	
20	Format	2	1=uncompressed PCM	
22	Channels	2	1=mono,2=stereo, more for other formats	
24	Sample Rate	4	44100, 16000, 8000 etc.	
28	Byte Rate	4	SampleRate * Channels * (BitsPerSample/8)	
32	Block Align	2	Channels * (BitsPerSample/8)	
34	BitsPerSample	2	8,16,24 or 32	
36	ID	4	The letters "data"	Data Section
40	Data Size	4	Size of the data below	
44	Data		The sound data	



Rest of sound data until end of data
 Format is first channel sample followed
 by second channel sample then back to first
 channel, etc. etc.
 Each sample will be BitsPerSample in size

RIFF - Wave



Steganography

Audio steganography techniques:

- Least significant bit encoding;
- Dephasing techniques;
- Hiding a message by adding an undetected echo for the human ear;

Is what you expected?

This was [Section 2.3 – Sound](#)

1. Features (frequency, wavelength, amplitude, intensity, speed, direction, pitch)
2. Codecs (MP3, FLAC)
3. Compression
4. File formats
5. Internal structure WAV – RIFF
6. Steganography

2.4 Video



Video

- Is the most complex element of multimedia field
- Is based on fast displaying of static images (frames)
- Human eye doesn't detect additional frames above a number of 12-15 frame per second
- For a video sequence of **720*486 pixels** running with **30 frames** per second, the computer must process around **21 MB per second**

Video

- Digital representation comes with high fidelity and numerous ways of processing digital content.
- The most important aspect is how video frames are displayed

Video

- Computer screens and some video devices uses a video signal based on three main colours of the RGB format
- The TV band and the majority of video systems use a compound signal in which brightness, hue and frame synchronization are all combined into a one signal

Video

- ways of displaying video signal:
 - interlaced video signal is used in analogic systems; two different sets of lines alternate, odds and evens, running video sequences with 25 frames/second;
 - on digital systems the progressive video signal is used, lines being displayed one after another

Video

- Interlaced – the image is fully displayed but then it gets gradually clearer until its final shape
- Progressive – means that the image is rendered line by line in a progressive manner until full clarity

Video

Interlaced



<http://justsayyes.wordpress.com/2007/06/28/demonstration-of-an-interlaced-video-and-deinterlaced-video/>

Video

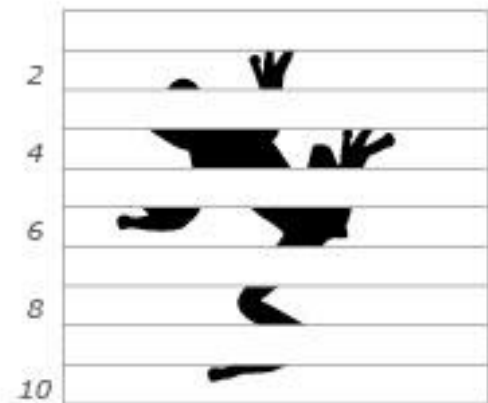
Progressive: full frame at once



Interlaced: frame split up into two fields



Field 1. odd lines



Field 2. even lines

Video

- How colours are rendered:
 - Analog : brightness and hue
 - Digital : RGB signal
- Displaying the image:
 - Analog : interlaced, small refresh rate
 - Digital : progressive, high fidelity
- Resolution:
 - Analog : PAL -625 lines, NTSC – 525 lines
 - Digital : PC – starting with 480.

Video

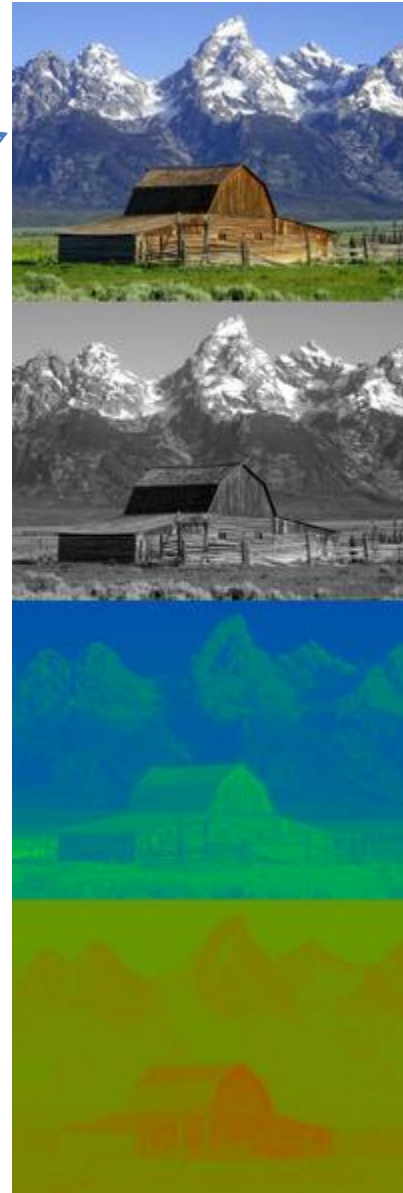
Three types of video signals:

- Component-VIDEO:
 - For analogue YUV : (Y-brightness, UV-hue)
 - For digital : RGB
- Composite-VIDEO: all the components are mixed together;
- S(separated)-VIDEO: different signal for each component.

Analogous video

YUV

YUV was used for sending color signal in an infrastructure that uses only black and white images.



[Wiki]

Video - analogue

Composite – Video

- RCA cables
- DVD Player – TV, Camera Video - TV



S – Video

- Uses S-Video cables
- PC – TV, Camera Video - TV



Video - analogue

- TV analogous standards for encoding video content and producing a electronic signal:
 - **PAL**
 - **SECAM**
 - **NTSC**

Video

- NTSC (National Television Standards Committee)
- Released in 1953
- A single video frame made up of 525 horizontal lines and running with a rate of almost 30 frames per second;
- 45 lines are used for image synchronization and the rest for the actual video content
- Displaying a frame is made by combining two fields of 240 lines each one odd and one even

Video

SECAM (Sequential Colour and Memory)

- Released in the 60s, a system with 525 lines
- The frame rate is around 25 frames per second

PAL (Phase Alternate Line)

- Released in 1966 in the European market
- Uses 625 lines per image with a 25 frames per second
- Only 576 lines are considered active and used for image representation, the other 49 are reserved for synchronization

Video

HDTV (High Definition Television)

- Programmable system based on digital television
- High resolutions that can be used also on computers

Video

HDV (High Definition Video)

- High resolution: 1,280×720 pixels (720p) or 1,920×1,080 pixels or 4k or 8k

Video mode	Dimension	Pixels	Display method	Frame rate (Hz)
720p	1,280×720	921,600	Progressive	23.976, 24, 25, 29.97, 30, 50, 59.94, 60, 72
1080i	1,920×1,080	2,073,600	Interlaced	25 (50 fields/s), 29.97 (59.94 fields/s), 30 (60 fields/s)
1080p	1,920×1,080	2,073,600	Progressive	23.976, 24, 25, 29.97, 30, 50, 59.94, 60

Video compression

- Makes use of redundant frames;
- Frame redundancy :
 - Spatial = intra-frame;
 - Temporal = inter-frame;
- The objective is to preserve the meaning of the each video frame;

Video compression

- Lossy compression
- Lossless compression

Real time video compression algorithms:

- JPEG, MPEG, DVI, M-JPEG;
- Based on the types of frame redundancy
- Compresses video content with rates from 50:1 up to 200:1

Video

The MPEG video standard:

- MPEG1 → video compression, asynchronous sound;
- MPEG2 → digital video used in TV transmissions
- MPEG3 → high resolution for digital television
- MPEG4 → used for video streaming over the Internet

The algorithm is a hybrid one which uses:

- Spectral analysis which by means of a cosine discrete transformation identifies repeatability;
- Hoffman encoding
- Differential coding identifies the differences between frames in order to display them, uses data from previous frames for displaying the next ones;
- Predictive coding analysis the elements that are changing between frames;

Video

Lossless video codecs:

- AVIzlib
- CamStudio GZIP
- CorePNG
- FastCodec
- MSU Lossless Video Codec
- PICVideo
- TSCC TechSmith Screen Capture Codec
- ZMBV (Zip Motion Block Video) Codec
- JPEG 2000
- YULS

Video

Lossy video codecs:

- Cinepak
- H.264 - MPEG-4 AVC (Advanced Video Coding)
sauMPEG-4 Part 10, approved for Blu-ray
- MJPEG
- JPEG 2000 intra frame video codec MPEG-1 Video
(MPEG-1 Part 2)
- MPEG-1 Video (MPEG-1 Part 2)
 - Cinema Craft Encoder
 - FFmpeg

Video

Lossless video codecs:

- MPEG-2 Video (MPEG-2 Part 2) -H.262
 - Cinema Craft Encoder
 - FFmpeg
 - InterVideo Video Decoder
 - Ligos LSX MPEG-2
 - MainConcept MPEG-2
 - TMPGEnc
- MPEG-4 ASP (Advanced Simple Profile) or MPEG-4 Part 2
 - DivX
 - FFmpeg MPEG-4
 - HDX4
 - Nero Digital
 - Xvid

Video

- RealVideo
- Snow Wavelet Codec
- Sorenson Video, Sorenson Spark
- Tarkin
- Ffmpeg
- TruDef high definition fractal video codec
- VC-1 (SMPTE standard, subset of Windows Media Video)
- VC-3 SMPTE standard
 - Avid DNxHD
 - FFmpeg
- Windows Media Video (WMV)

Video – storage media

	Blu-ray Disc	HD DVD	DVD
Capacity	25 GB	15 GB	4.7 GB
Maximum bit rate – data transfer	53.95 Mbit/s	36.55 Mbit/s	11.08 Mbit/s
Maximum bitrate – audio + video + subtitles	48 Mbit/s	30.24 Mbit/s	10.08 Mbit/s
Maximum bitrate – Video	40 Mbit/s	29.4 Mbit/s	9.8 Mbit/s
Video resolution	1920 x 1080	1920 x 1080	720×480 (NTSC) 720×576 (PAL)
Frame rate	24 p, 50/60i	24p, 50/60i	24 p, 50/60 i

Video Processing

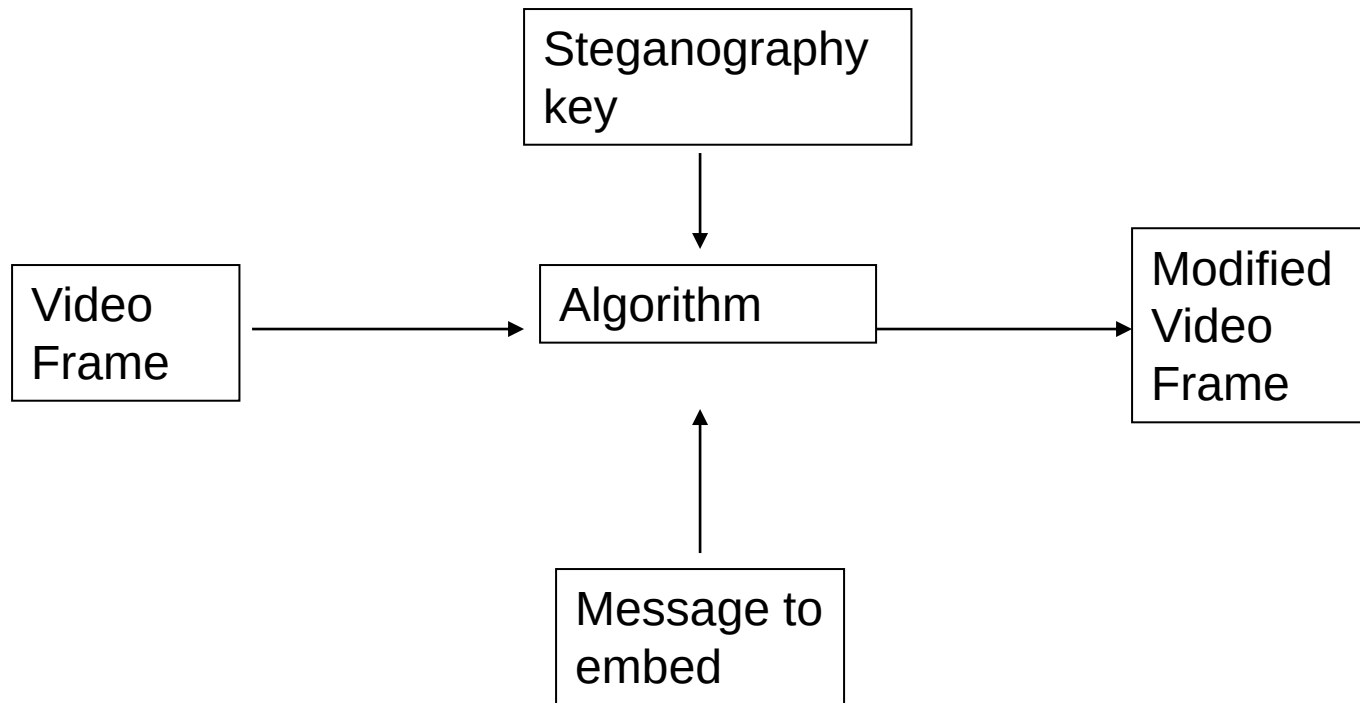
Direct access to video frames:

- TIME CODE (each frame has a unique BCD number (Binary Code Decimal), which gives the hour, minute, second and millisecond;
- FRAME CODE (based on the frame indexing mechanisms, each frame has a ordered positive number;

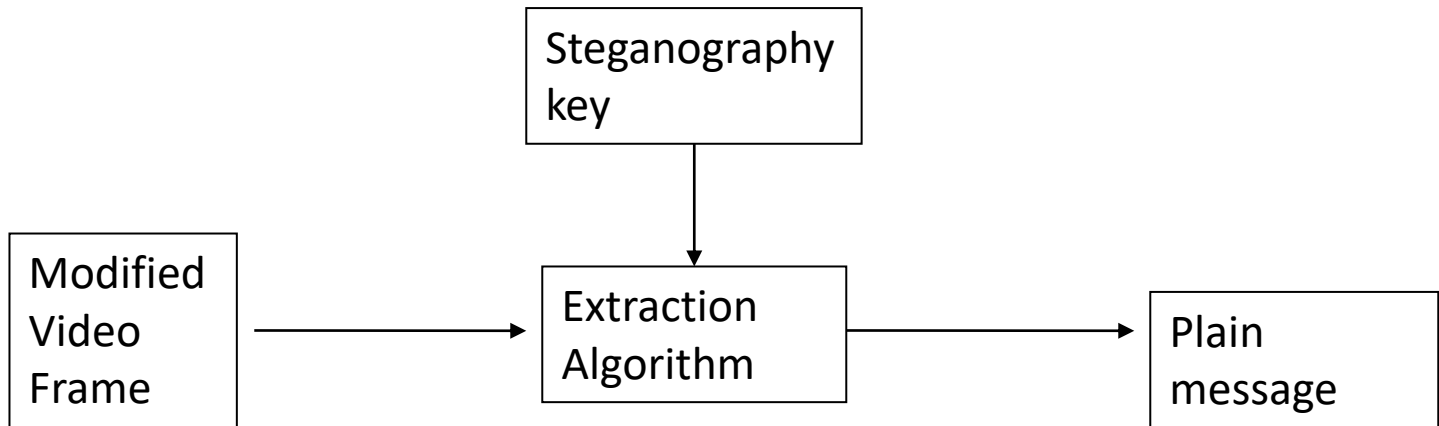
Steganography techniques

- Conventional oriented on images such as: intra frame, spatial domain, transformed domain;
- Recent techniques based on motion vectors and compression algorithms;

Steganography – embedding



Steganography - extracting



Watermarking

- Protects the authenticity and authorship by using algorithms that prevent the restoration of digital content to its natural state;
- Assures digital content integrity by using fragile watermarking techniques;

Is what you expected?

This was [Section 2.4 – Video](#)

Concept

Characteristics (speed, resolution)

Representation (analogic, digital)

Color representation (RGB, YUV)

Ways of displaying images:

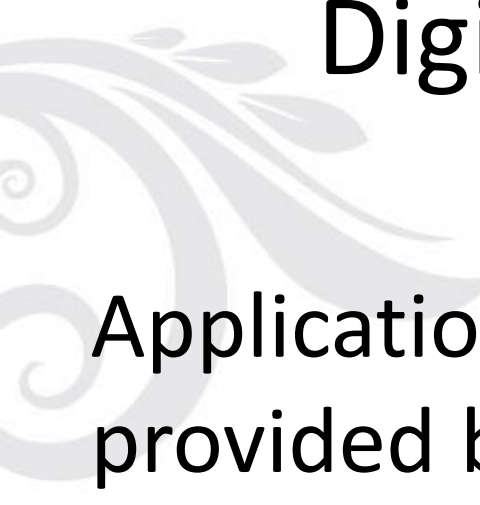
- interlaced
- progressive

Compression type (spatial, temporal)

Video codecs (DivX, Xvid, Ffmpeg,
Cinepak)

2.6 Digital Rights Management





Digital Rights Management

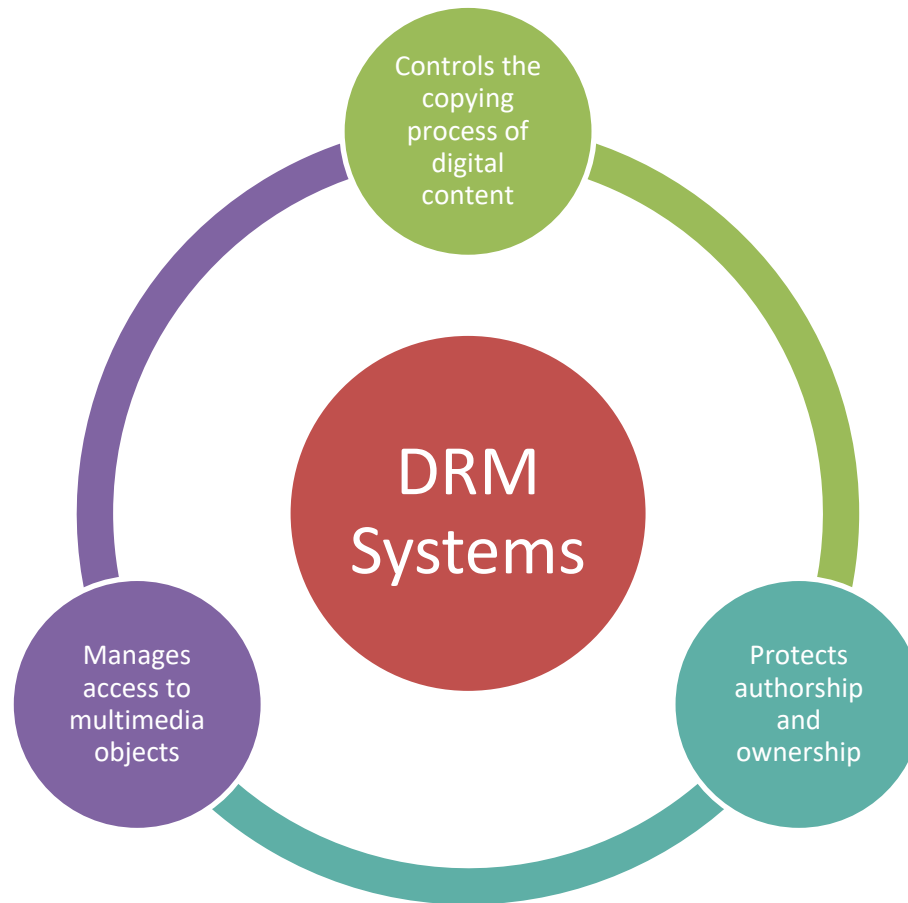
Application and enforcement of regulations provided by law for digital content by use of computer systems.

Digital Rights Management

Objectives:

- protecting the digital content;
- protecting ownership and authorship;
- allowing temporary access to digital content for a limited period of time;
- restrict access to certain features;

Digital Rights Management



Digital content protection

Digital Rights Management Protection

Technological means

Contractual
means

Access
control

Usage
control

Rights
expression

DRM Roles

From a technological perspective, a DRM system must have the following two distinctive roles that must complement each other, as follows:

- access control – control access to digital content and allow only authorized users to make use of it;
- usage control – protect rights of the copyright owner by limiting what an authorized user can do with the digital content by controlling the entire distribution chain of multimedia material;

DRM Methods

The role of a DRM system is to protect or limit the access to the digital content by unauthorized consumers by:

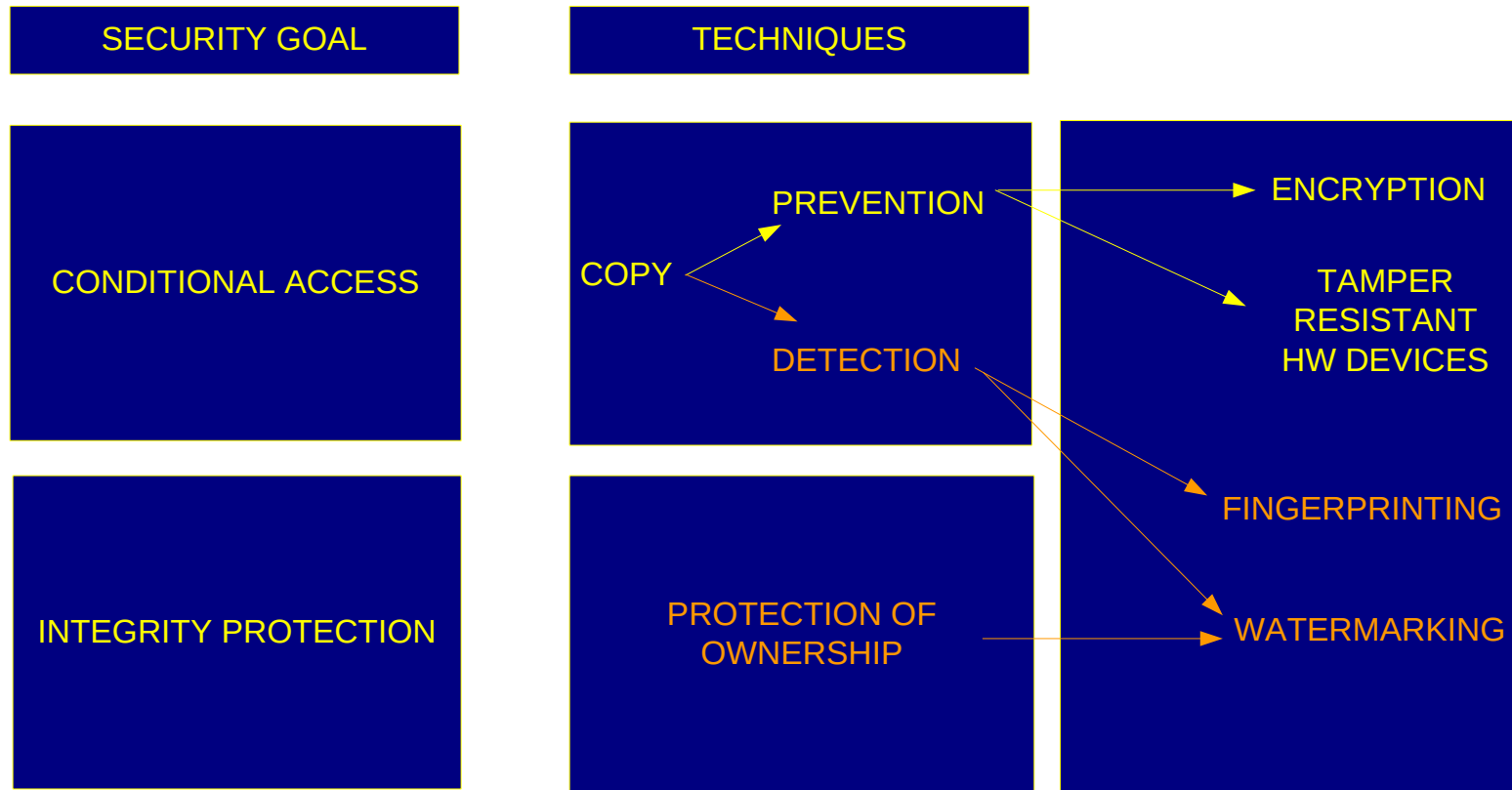
- limit the access by using encryption based digital containers;
- rights lockers architectures used to allow users to access their own digital materials from different types of devices;
- copy generation management systems, or CGMS, limits the number of copies that a user can make from its digital content; after this limit the content becomes unusable.

DRM Attacks

DRM system face two major types of attacks that can allow a malicious user to access digital content without permission:

- attack for finding the encryption key; every user that has access to the encryption key is allowed to view and manipulate the digital content;
- unencrypted content capturing; for this attack users must exploit flaws in the standardization procedure of a DRM system.

Digital Rights Management



2.5 Digital signature





Digital signature

Digital signature is a bi-univocal connection between a digital document and its author used in order to reflect a security feature called non-repudiation.



Digital signature

Objectives:

- ensures the authenticity of signed documents;
- detects changes by maintaining the integrity;
- identifies the signer uniquely - non-repudiation;

Digital signature

Creating public and private keys:

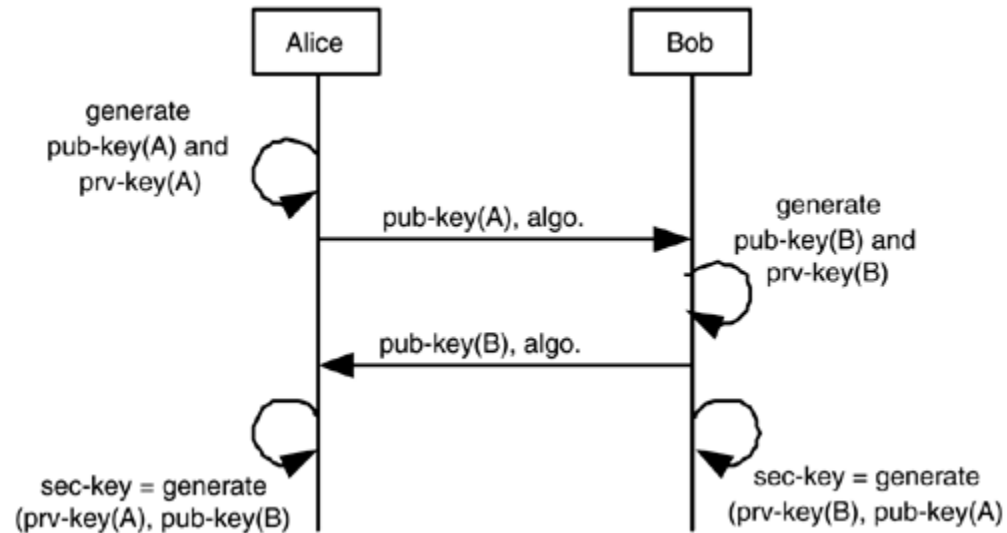
```
keytool -genkey -alias sender -keystore  
testkeystore.ks -keyalg RSA/EC
```

```
keytool -genkey -alias recev -keystore  
testkeystore.ks -keyalg RSA/EC
```

Jpeg Digital signature

1. Hashing the content of the image
2. Encrypting the hash with a secret symmetric key
3. Signing the encrypted hash with the private key of the sender
4. Attaching the signature as an EXIF meta-data node to the Jpeg file
5. (Virtual) Send the image to another user
6. Hashing the content of the received image
7. Encrypt the hash with the same secret symmetric key
8. Verify the signature with the public key of the sender

KeyAgreement





ISM Issues Summary
for easy sharing

Section Conclusions

2.1 Security in Multimedia

2.2 Image

2.3 Sound

2.4 Video

2.5 Digital signature

2.6 Digital Rights Management

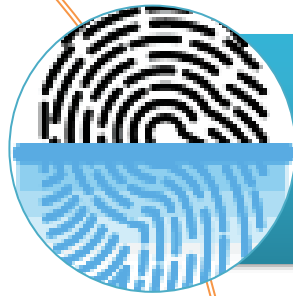


Face detection / recognition, Biometrics Systems

Biometry Aspects



Security Triade



What we are



What we know

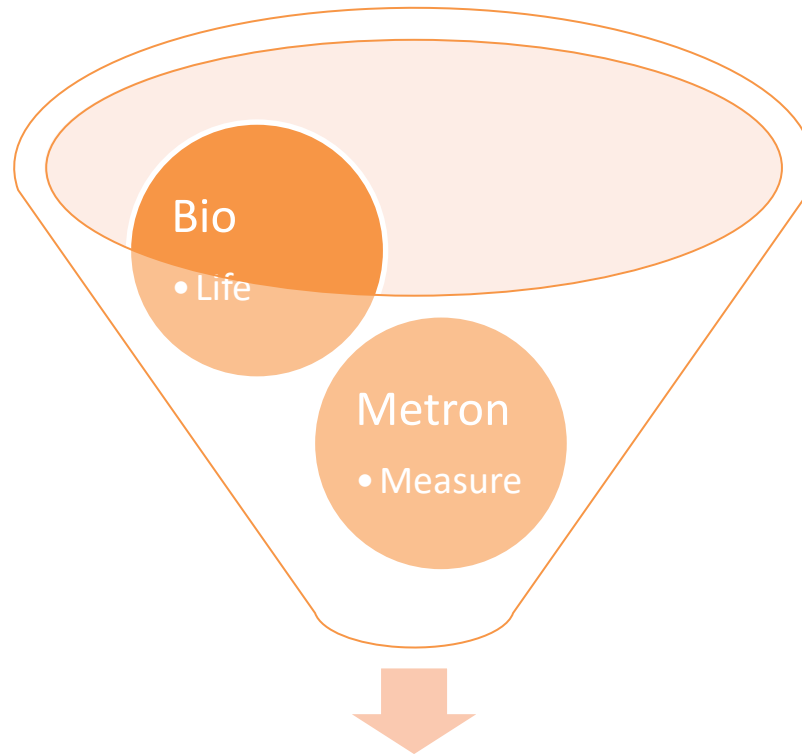


What we have

Biometry agenda

- Biometrics
- Performance Metrics
- Face recognition
- Biometric Applications

Etymology



Biometrics

Biometric systems

Automated measurement of physical and/or behavioral characteristics to determine or authenticate identity.

History

Babylonian merchants used fingerprints to settle business transactions that were recorded in clay tablets



Chinese used fingerprints and footprints to differentiate children



Early Egyptian used traders identified by their physical descriptions

Timeline

1858	1936	1960	1974-1999	1991	2002-2005	2005-2016
•Modern biometrics •First capture of a hand image for identification purposes by Sir William Herschel who recorded the handprint on the back of a contract for each worker to distinguish employees when it came to pay day	•Iris pattern proposed by ophthalmologist Frank Burch to recognize individuals	•A classification system for fingerprints •Automated signature recognition research begins •First work on facial recognition	•Hand geometry systems become available •Biometric commercial standard is published •Biometric Consortium established in US Gov.	•First patent on iris patterns recognition	•Technical Committee on Biometrics, JTC •Open Source Biometrics Libraries	•Behavioral biometrics

A Timeline of Biometric Authentication



First fingerprints recorded

There is evidence that fingerprints were used as a person's mark as early as 500 B.C. Babylonian business transactions are recorded in clay tablets that include fingerprints.

500BC

1858

**First systematic capture of
hand images for
identification purposes is
recorded in India**

Sir William Herschel recorded a handprint on the back of a contract for each worker to distinguish employees when it came to pay day.





Biometrics used to identify criminals

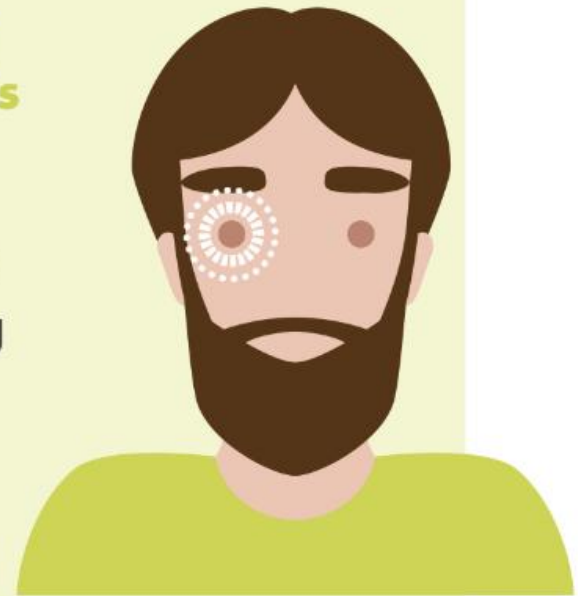
A French police officer, Bertillon, used body measurement taken of specific anatomical characteristics to identify reoffending criminals.

1888

1936

Concept of using the iris pattern for identification is proposed

Ophthalmologist Frank Burch proposed the concept of using iris patterns as a method to recognise an individual, a major breakthrough in biometric authentication.





First voice recognition tool launched

Dragon launches Dragon Dictate, the first speech recognition product for consumers.

1990

1996

Hand geometry is implemented at the Olympic Games

At the 1996 Atlanta Olympic Games, hand geometry systems were implemented to control and protect physical access to the Olympic Village.



2010's

Mobile biometrics become the norm



This decade included several developments in mobile biometrics; the launch of Apple's Siri paved the way for voice recognition and digital personal assistants on smartphones, finger print scanners became commonplace, and facial recognition became the norm.

The developments in mobile biometrics have gone a long way to increase social acceptance for biometrics as a means of authentication.

2020's

Biometrics extend beyond mobile

Biometric solutions have continued to evolve, and use of multi-biometric identification has become a part of our day to day lives.



The first commercial biometric payment card offers customers the option to use their fingerprints to authorise payments without the need to enter a PIN code – providing greater security.

Biometric pathways are now part of airport travel. They provide passengers with a smooth and intuitive, yet secure travel experience at every checkpoint in an airport, from self-check-in to boarding.

Biometrics use cases

- Law enforcement and public security
- Military
- Border control, travelling, and migration
- Healthcare and insurance sector
- Civil identity, voter registration
- Physical access control
- Commercial applications

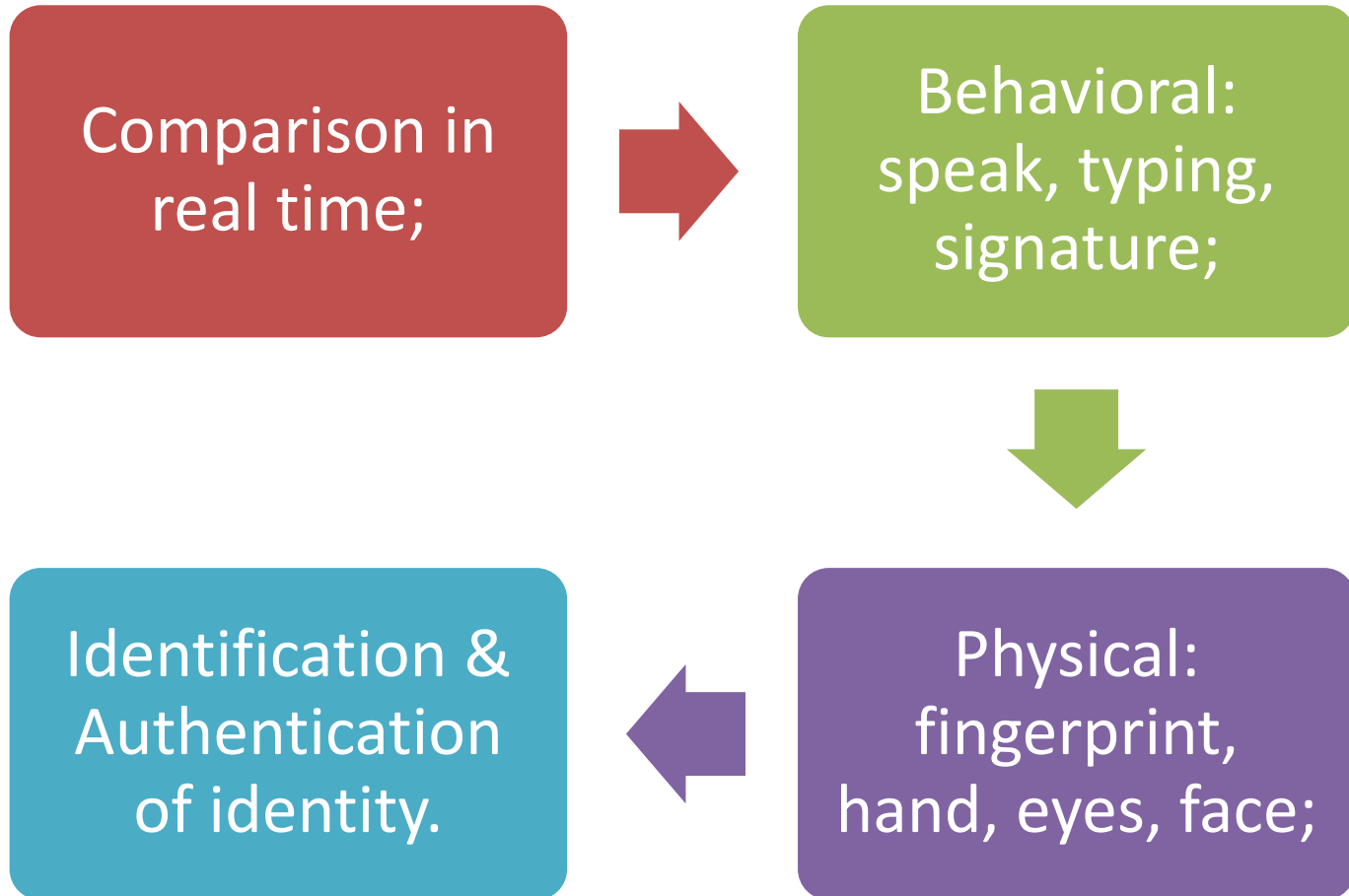
Types of Biometrics



Types of biometrics

- Fingerprint
- Face Recognition
- Hand Geometry
- Iris Scan
- Voice Scan
- Signature
- Retina Scan
- Infrared Face and Body Parts
- Keystroke Dynamics
- Gait
- Odor
- Ear
- DNA Matching

Biometrics



Identification vs. Authentication

Identification	Authentication
It determines the identity of the person.	It determines whether the person is indeed who he claims to be.
No identity claim Many-to-one mapping. Cost of computation $\propto$ number of record of users.	Identity claim from the user One-to-one mapping. The cost of computation is independent of the number of records of users.
Captured biometric signatures come from a set of known biometric feature stored in the system.	Captured biometric signatures may be unknown to the system.

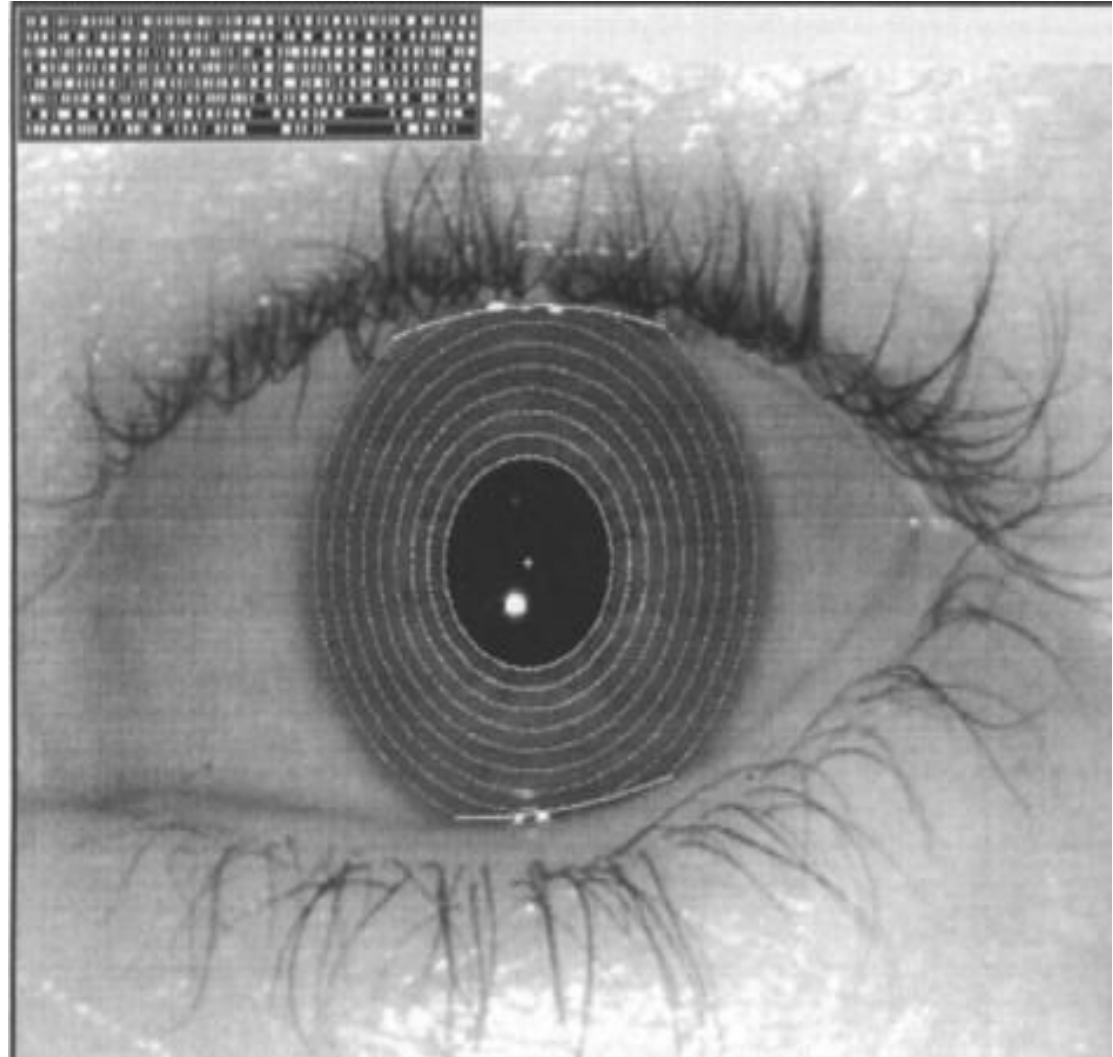
Fingerprint

- Crossover
- Core
- Bifurcations
- Ridge ending
- Island
- Delta
- Pore



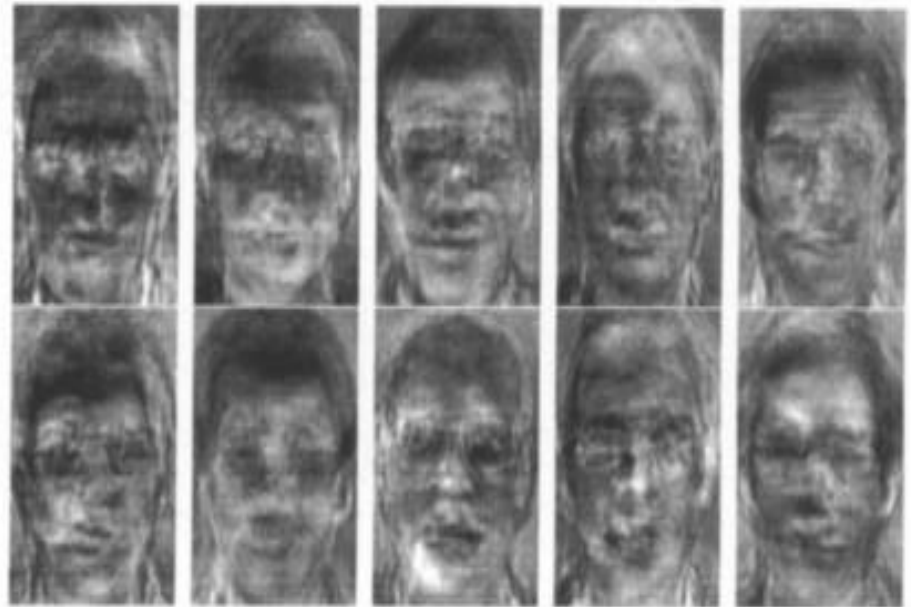
Iris Code

- Ridges (rings)
- Furrows
- Striations (freckles)



Face Recognition

Distance between the eyes;
Distance between the eyes and nose ridge;
Angle of a cheek;
Slope of the nose;
Facial Temperatures;



Biometrics Characteristics

Universality

Uniqueness

Permanence

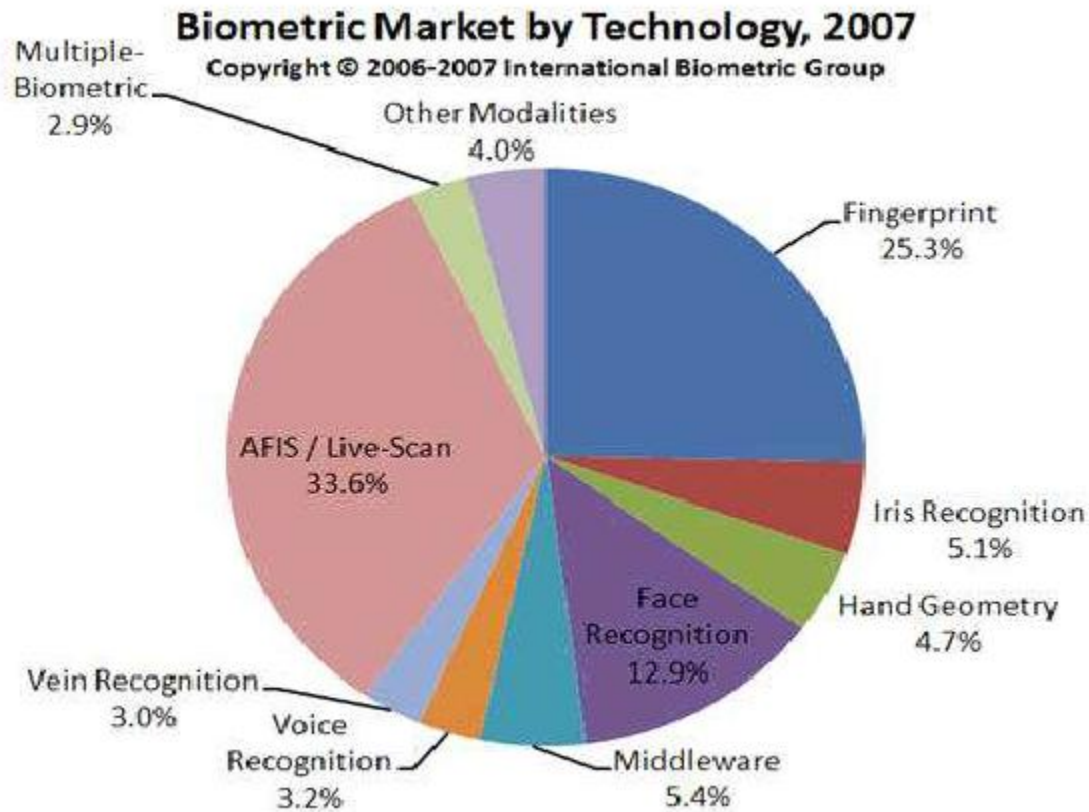
Collectability

Performance

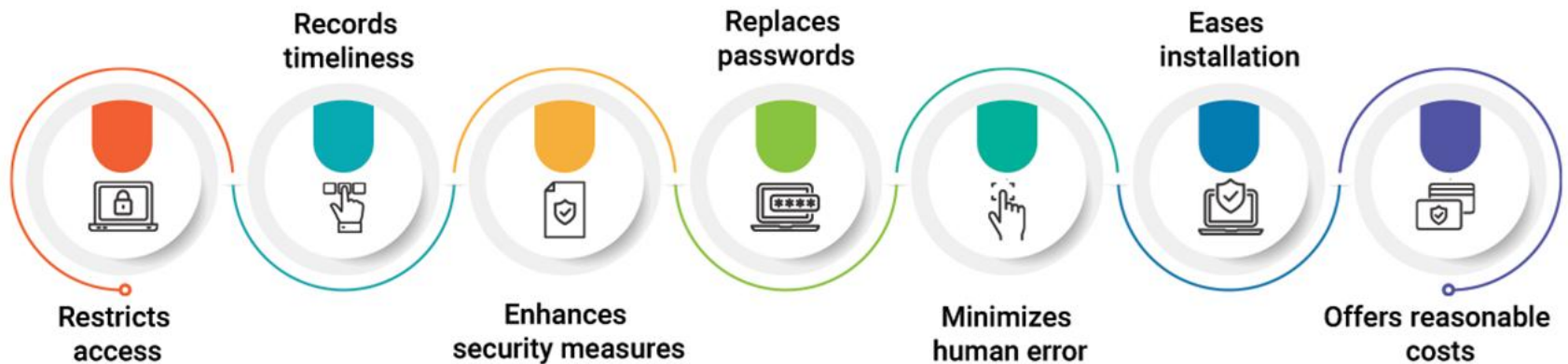
User's Acceptability

Robustness against Circumvention

Biometric Market



Biometric authentication



Face recognition

- Distinguishable landmarks as nodal points (about 80), 14-22 used
- Distance between eyes
- Width of nose
- Depth of eye sockets
- Cheekbones
- Chin
- Jawline



Face recognition

Advantages

- Most promising biometric for overcrowded places (airports)
- Highest user-acceptance

Problems

- Surveillance: animated expression, sunglasses, aging, hairstyle, weight gain/loss
 - Dynamic, Uncontrolled face identification in cluttered background
 - Identical twins
 - Printed masks
- 

Methods for Face Recognition

- Eigenfaces
- Fisherfaces
- Neural networks
- Circular Local Binary Pattern

Eigenfaces

- Compute a "distance" between the new image and each of the example faces
- Select the example image that's closest to the new one as the most likely known person
- If the distance to that face image is above a threshold, "recognize" the image as that person, otherwise, classify the face as an "unknown" person

Biometric verification

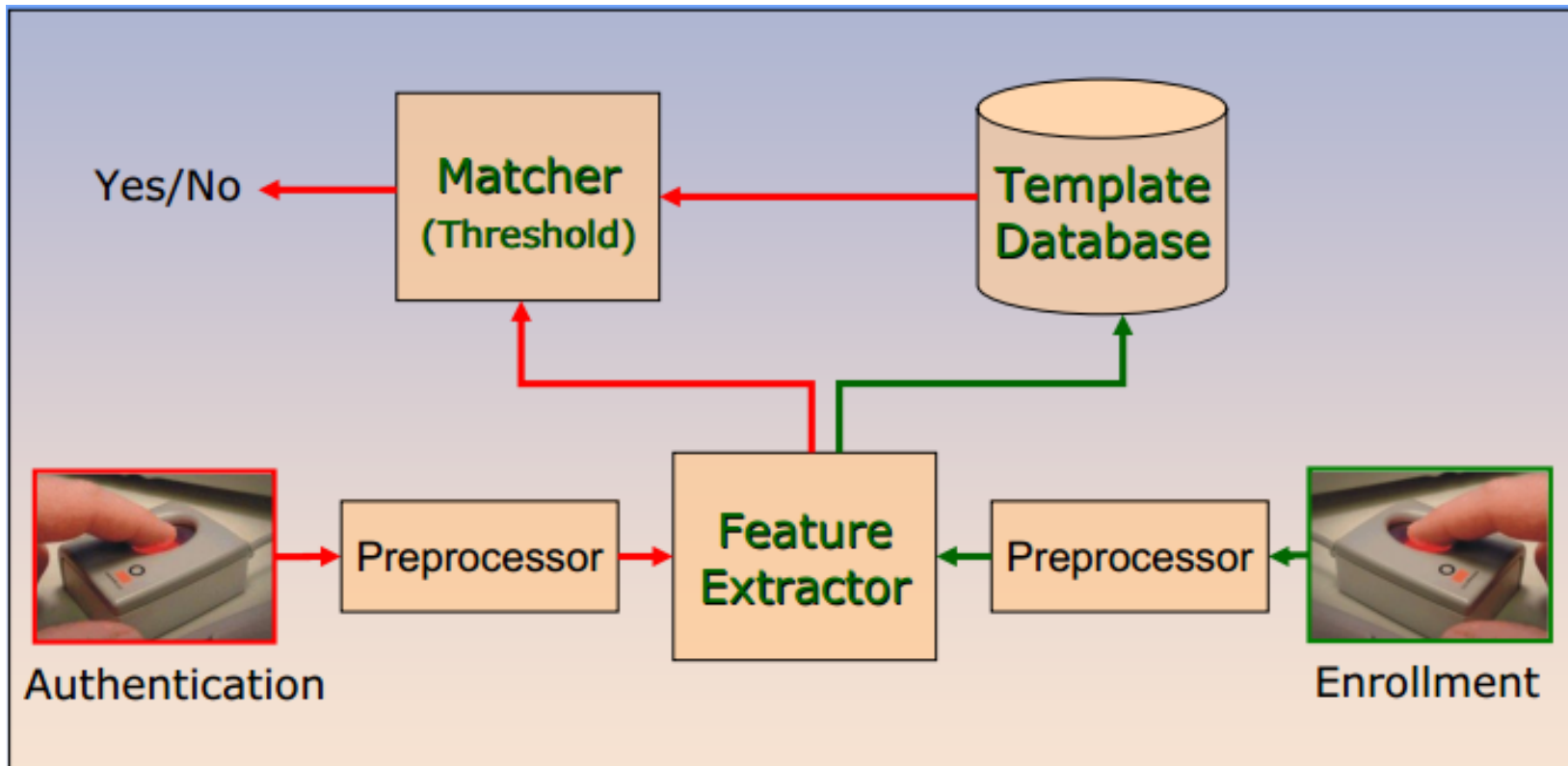
- I. The claimed identity I has feature vector fv from his biometric data, saying it is class $C1$;
- II. Task is to determine if (I, fv) belongs to class $C1$ or $C2$
- III. Decision $D(fv, fi)$ by calculating the similarity

$$(I, fv) \in \begin{cases} C1, & D(fv, fi) \leq t \\ C2, & otherwise \end{cases}$$

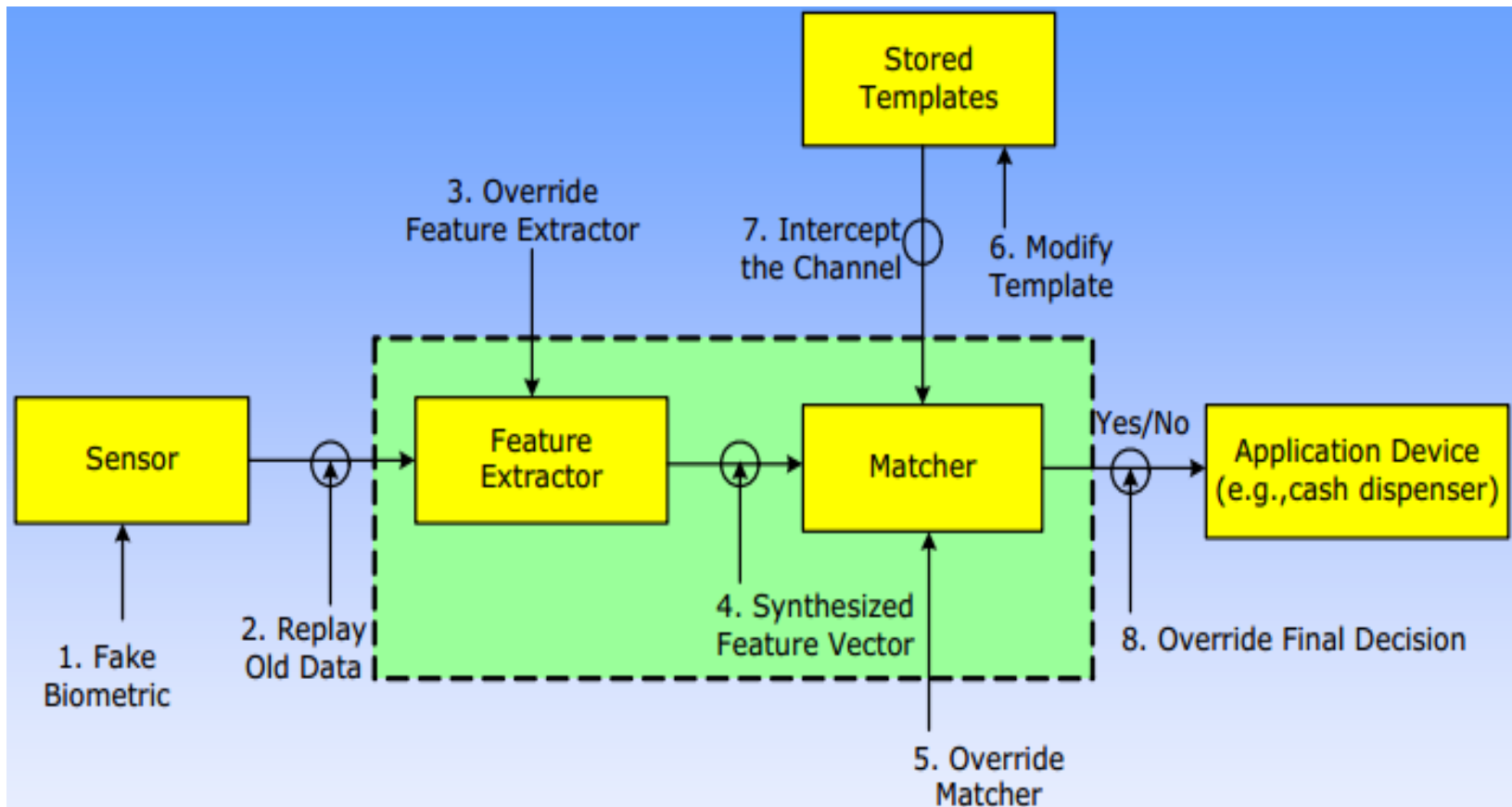
Biometric system

- Biometric acquisition?
- Feature extraction?
- Data storage
- Matching against the template in database
- Decision implementation

Biometrics procedure



Biometrics attacks



Performance Metrics

- Confusion Matrix
- FAR and FRR
- Distributed Analysis
- Threshold Analysis
- Receiver Operating Curve

Optimal biometrics

More accurate.

Less expensive.

Faster.

More limited in
function.

Requires more effort
by user.

Confusion matrix

- In the confusion matrix the rows represent the target classes and the columns the output classes for a testing target data set. The diagonal cells in each table show the number of cases that were correctly classified, and the off-diagonal cells show the misclassified cases.

- The confusion matrix is an $m \times m$, where m is the number of classes to be predicted. For binary classification problems, the number of classes is 2, thus the confusion matrix will have 2 rows and columns.

- It is needed to choose a decision threshold τ in order to label the instances as positives or negatives. If the probability assigned to the instance by the classifier is lower than τ , it is labeled as positive and if the probability is greater than the decision threshold, it is labeled as negative.

- True positives (TP), which are the instances that are positives and are classified as positives.
- False positives (FP), which are the instances that are negatives and are classified as positives.
- False negatives (FN), which are the instances that are positives and are classified as negatives.
- True negatives (TN), which are the instances that are negatives and are classified as negatives.

Confusion Matrix

		Recognized as							
True ID		Reject	A	B	C	D	E	F	ACCURACY %
	A	6	168	1	8	17	8	5	78
	B	7	1	154	1	10	1	0	88
	C	17	1	0	77	0	10	1	72
	D	13	5	2	2	215	7	1	87
	E	22	1	0	3	5	64	0	67
	F	2	2	1	9	3	11	139	83

Classification accuracy: Ratio of instances correctly classified. An accuracy of 100% means that the measured values are exactly the same as the given values.

$$\textit{classification_accuracy} = \frac{\textit{true_positives} + \textit{true_negatives}}{\textit{total_instances}}$$

Error rate: Ratio of instances misclassified

$$error_rate = \frac{false_positives + false_negatives}{total_instances}$$

Sensitivity, or true positive rate: Proportion of actual positive which are predicted positive. For example, a sensitivity of 100% means that the test recognizes all actual positives, i.e., all sick people are recognized as being ill.

$$\textit{sensitivity} = \frac{\textit{true_positives}}{\textit{positive_instances}}$$

Specificity, or true negative rate: Proportion of actual negative which are predicted negative,.

$$specificity = \frac{true_negatives}{negative_instances}$$

Positive likelihood: Likelihood that a predicted positive is an actual positive.

$$positive_likelihood = \frac{sensitivity}{1 - specificity}$$

Negative likelihood: Likelihood that a predicted negative is an actual negative.

$$\text{negative_likelihood} = \frac{1 - \text{sensitivity}}{\text{specificity}}$$

False Acceptance Rate (FAR)

$$FAR = \frac{\text{Total False Acceptance}}{\text{Total Attempts}}$$

False Rejection Rate (FRR)

$$FRR = \frac{\text{Total False Rejection}}{\text{Total Attempts}}$$

FTE & ATV

$$FTE = \frac{\text{Total Rejections}}{\text{Total Attempts}}$$

$$ATV = (1-FTE) (1-FRR)$$

Biometrics Distribution

Biometrics	Market Share
Fingerprint	34%
Hand Geometry	26%
Face Recognition	15%
Voice Authentication	11%
Iris Recognition	9%

OpenCV

- <http://www.opencv.org>
- Platforms: C++ / Java / Android / CUDA / Clojure



Steganography, Zoom, Compression, Sound, Digital Signature

Hands-On Exercises





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